

A Separation of Uncertainty Attitudes and Belief Updating

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26th September 2026

Abstract

This paper proposes a natural and tractable updating rule for Choquet expected utility with probabilistic sophistication, grounded in Bayesian updating. A probabilistically sophisticated decision maker expresses beliefs by a subjective probability measure, which serves as a Bayesian prior and is updated to a posterior according to Bayes' rule when new information becomes available. Under certain conditions, a non-additive capacity can be represented as a transformation of an additive probability measure. Leveraging this representation, beliefs are updated by applying Bayes' rule to the underlying probability measure, with the updated capacity derived by applying the same transformation to posterior probabilities. We provide an axiomatisation of this updating procedure and develop a simple and intuitive updating rule that neatly separates beliefs from attitudes towards uncertainty, with beliefs evolving in a Bayesian manner and uncertainty attitudes remaining unchanged.

Keywords: Choquet expected utility, probabilistic sophistication, Bayesian updating

1 Introduction

In reality, nearly all economic decisions involve unknown probabilities to a varying extent. Even though the probabilities of events are not known, the decision maker (DM) may hold beliefs about the likelihood of events, which can be represented by a subjective probability distribution. In a seminal contribution to the theory of decision making under uncertainty, Savage (1954) shows that, under a set of behavioural axioms, we can deduce a unique subjective probability distribution over events with unknown probabilities from the DM's choice behaviour. Under these axioms, the DM behaves as if maximising an expected utility functional defined over state-contingent outcomes, where expectations are taken with respect to the inferred subjective probability distribution.

Savage's subjective expected utility (SEU) theory provides an appealing extension of the expected utility framework to settings in which the probabilities of events are unknown. Within this framework, uncertainty is accommodated by modelling agents who satisfy the Savage's axioms as expected utility maximisers endowed with subjective beliefs over states of the world. However, the behavioural assumptions underpinning the existence of a subjective probability distribution have been widely challenged by empirical evidence. In many situations involving uncertainty, people do not appear to hold well-defined subjective probabilities. Experimental investigations have shown that SEU fails to adequately capture the DM's behaviour under uncertainty as it cannot account for violations such as the Allais paradox (Allais, 1953) and the Ellsberg paradox (Ellsberg, 1961). These findings have motivated the development of alternative models of decision making that relax or depart from the Savage's axioms in order to better accommodate observed behaviour under ambiguity.

If SEU is inadequate, how should ambiguity be modelled? One of the most prominent approaches is Choquet expected utility (CEU), which rests on rigorous axiomatic foundations developed by Gilboa (1987) and Schmeidler (1989) and subsequently refined in the literature (Wakker, 1989; Sarin and Wakker, 1992; Abdellaoui and Wakker, 2005). In the CEU framework, beliefs are represented by a *capacity*, i.e. a non-additive set function, which relaxes the additivity requirement imposed on subjective probabilities under SEU. This feature allows CEU to capture ambiguity by permitting the DMs' beliefs to deviate from probabil-

istic additivity.

The use of non-additive beliefs is psychologically plausible. A substantial experimental literature suggests that individuals perceive likelihood in a non-linear manner, a phenomenon that is usually more pronounced under uncertainty than under risk (Currim and Sarin, 1989; Fellner, 1961; Tversky and Wakker, 1998; Weber, 1994). Moreover, CEU embodies the principle of *rank dependence*, whereby the decision weight attached to an event depends not only on the event itself but also on the relative ranking of the outcome it yields compared to outcomes associated with other events. In this sense, CEU generalises SEU by allowing the decision weight of an event to depend on its ranking position rather than solely on the event itself.

In addition, CEU has generated fruitful applications in decision theory for several reasons. First, it is mathematically sound and avoids implausible behavioural implications, such as violations of stochastic dominance (Fishburn, 1978). Second, CEU can accommodate a wide range of empirically observed violations of expected utility theory (Abdellaoui and Munier, 1999; Birnbaum and McIntosh, 1996; Bleichrodt and Pinto, 2000; Gonzalez and Wu, 1999; Harless and Camerer, 1994; Tversky and Fox, 1995). Third, CEU is intuitively appealing: its underlying concepts provide economically meaningful insights into behaviour under ambiguity and offer a richer descriptive account of how individuals evaluate uncertain prospects (Diecidue and Wakker, 2001).

The analysis so far has focused on static decision problems under SEU and CEU. To extend models with non-additive beliefs to sequential or dynamic settings, it is necessary to specify how preferences—and in particular beliefs—are updated as new information becomes available. In SEU, the almost universally applied rule is Bayesian updating. The issue of updating ambiguous beliefs has been thoroughly studied as well. In contrast to probability theory, there are various approaches to updating Choquet beliefs in extant literature. Among them, there are three well-known ways to update capacities: the *naïve Bayes' updating rule* or *optimistic updating rule* (see Gilboa and Schmeidler, 1993), the *Dempster-Shafer updating rule* or *pessimistic updating rule* (see Dempster, 1968; Shafer, 1976) and the *generalised/full Bayesian updating rule* or *Fagin-Halpern updating rule* (see Dempster, 1967; Fagin and Halpern, 1991; Walley, 1991; Jaffray, 1992). We will provide the formal definitions of

these three updating rules in Section 4. In addition, various axiomatic-based updating rules have been proposed for CEU (Gilboa and Schmeidler, 1993; Eichberger et al., 2007). However, if we apply these three updating rules to piecewise additive Choquet expected utility (PACE utility) developed by Webb (2015), a continuous version of NEO-expected utility (NEO-EU) will become discontinuous. More details will be discussed in Section 4.

Can we find a simple updating rule for capacities? The answer is yes, provided that certain conditions are satisfied. In particular, a non-additive capacity may, in some cases, be expressed as a strictly increasing transformation of a probability measure. When this holds, the DM is said to be a *probabilistically sophisticated* CEU maximiser, with non-additivity capturing attitudes toward uncertainty. Further discussion of probabilistic sophistication is provided to Section 6. Building on this special case, this paper proposes a natural and intuitive updating procedure for capacities by the application of Bayes' rule in situations where CEU admits an equivalent rank-dependent utility representation. As shown by Wakker (1990), under stochastic dominance, CEU and rank-dependent utility are identical.

From this perspective, this paper essentially characterises a simple updating rule for probabilistic sophistication in the context of CEU. If the capacity can be further expressed as a probability transformation, how do we update such a subjective probability distortion? Such a probabilistically sophisticated agent may quantify uncertainty using a prior probability measure and update this prior to a posterior in a Bayesian manner, while maximising CEU with respect to her probabilistic beliefs. The research questions addressed in this paper are summarised as follows:

1. How does a probabilistically sophisticated CEU maximiser update her beliefs upon receiving new information?
2. Under what conditions can a DM update her beliefs while keeping her attitudes towards uncertainty unchanged?
3. What are the behavioural implications (axioms) for preferences and how might these properties change (or not) in dynamic choice?

When modelling a DM's behaviour in a dynamic, ambiguous context, a major problem lies in the well-known precarious relationship between updating capacities, *consequential-*

ism and *dynamic consistency*. Consequentialism requires that only those outcomes on states which are still possible can matter for preference. Dynamic consistency requires that preferences after some event that is likely to occur remain consistent with ex-ante preferences. Early work by Ghirardato (2002) showed that together the three axioms (i.e. state independence, consequentialism and dynamic consistency) imply for a CEU maximiser that the utility function remains unchanged for the conditional preferences and the capacity is additive and updated by the Bayesian rule. State independence requires the ex-post preferences over outcomes to agree with the ex-ante preferences over outcomes. Formal definitions of these three axioms will be given in Section 4. Dominiak and Lefort (2011) point out that dynamic consistency and consequentialism cannot be satisfied simultaneously in a dynamic, ambiguous context. To axiomatise the updating rules of ambiguous beliefs, there are essentially two ways to proceed. Either one abandons consequentialism or gives up dynamic consistency. In the extant literature, Hanany and Klibanoff (2007), Siniscalchi (2011) and Sarin and Wakker (1998a) drop consequentialism and retain dynamic consistency, while Walley (1991), Gilboa and Schmeidler (1993), Pires (2002), Eichberger et al. (2007) and Eichberger et al. (2012) retain consequentialism and drop dynamic consistency. In particular, Eichberger et al. (2007) retain consequentialism and proposes a weaker version of dynamic consistency to axiomatise the full Bayesian updating rule for capacities. Inspired by the ideas of Eichberger et al. (2007) and Eichberger et al. (2012), we retain consequentialism in this paper.

Our aim of this paper is to develop a simple model to update Choquet beliefs and to provide axiomatic foundations in a dynamic framework for a CEU maximiser. In particular, a capacity is a probability transformation of an additive probability measure. Beliefs continue to be represented by probabilities, but we additionally allow each DM to exhibit an uncertainty attitude. Importantly, DMs need not hold precise or explicitly articulated beliefs; rather, their beliefs may be vague or imprecise, reflecting the inherent uncertainty of the decision environment. Based on these considerations, the idea underlying the updating procedure is as follows. Under certain conditions specified by Gilboa (1985) and Sarin and Wakker (1998b), capacities can be expressed as transformations of additive probability measures. We then update the additive probability measure by using the Bayes' rule.

We continue to update the corresponding capacity by setting the updated capacity of each event equal to the transformation of the updated additive probability of the same event.

The remainder of this paper is structured as follows. Section 2 deals with some preliminaries, namely, the framework of the model and some useful definitions; the Choquet integral. Section 3 introduces the simple model to update Choquet beliefs. Section 4 reviews the key axioms of SEU and Bayesian updating in a conditional decision problem and three popular updating rules for capacities. Section 5 studies the dynamic preference foundations and presents the main representation theorem. As this paper primarily characterises the updating rule for a probabilistically sophisticated DM in the context of CEU, Section 6 provides a more detailed discussion of probabilistic sophistication and uncertainty attitudes. Section 7 concludes and discusses the future work. All proofs are provided in the Appendix at the end of this paper.

2 Notation and Definitions

In this section, we present a setup for choice under uncertainty, following Sarin and Wakker's (1992, 1997) approach. As in Anscombe and Aumann's (1963) approach, a randomising device (e.g. a roulette wheel) is used to calibrate subjective probabilities and utilities. A key distinction, however, lies in the structure of outcomes. Whereas the Anscombe–Aumann framework allows each uncertain event to yield an arbitrary lottery, the Sarin–Wakker approach restricts outcomes to prizes (i.e. degenerate lotteries). This restriction avoids the need for a two-stage formulation and leads to a simpler, one-stage decision framework.

Building on this setup, we then introduce several definitions that will be useful for the subsequent analysis and present the CEU model of ambiguity. Our analytical framework adopts the one-stage formulation of Sarin and Wakker to derive CEU, which can be viewed as a unification of Schmeidler (1989) and Gilboa (1987).

2.1 Basic Setup of Decision Making under Uncertainty

States of the World. Introduced by Savage (1954), the model of choice under uncertainty begins with the idea of *states of the world*, or simply *states*. The *world* is the object that the

DM is concerned with. A *state*, denoted s , is a complete description of the world when all uncertainty has been resolved. We use the following notation:

$$\mathcal{S} = \{\dots, s, s', \dots\} \text{ is the set of states.}$$

The set of states $\mathcal{S}$ is assumed to contain every possible, or at least every relevant description of the world *after* the uncertainty has been resolved. Only one state can be true, and we say ‘the state s is true’ or ‘the state s obtains.’ Prior to the resolution of the uncertainty, the DM does not know which state will be true and has no influence over which state is realised. As an example, we may consider a horse race that will take place. Each horse corresponds to a distinct state of the world, and a state $s \in \mathcal{S}$ represents the event that horse s wins the race. We sometimes refer to $\mathcal{S}$ as the horse state space.

Events. Subsets of $\mathcal{S}$ are *events*, and we use the following notation:

$$\mathcal{E} = \{\dots, A, B, \dots, E, \dots\} \text{ is the set of events.}$$

We can refer to the state s belonging to an event A , $s \in A$. A^c denotes the complement of A . In this paper, we shall let $\mathcal{E}$ denote a σ -algebra of subsets of $\mathcal{S}$; that is, $\mathcal{E}$ contains $\mathcal{S}$, $A \in \mathcal{E}$ implies $A^c \in \mathcal{E}$, and $\mathcal{E}$ is closed under countable unions. Thus, $\mathcal{E}$ also contains $\emptyset$, and is closed under countable intersections.

Following Sarin and Wakker (1992, 1997), we identify two types of events: *ambiguous* (or *horse*) events, whose probabilities are not known, denoted by $\mathcal{H}$, and *risky* (or *roulette*) events, whose probabilities are known, denoted by $\mathcal{R}$. The set of ambiguous events, denoted by $\mathcal{H}$, consists of all subsets of an underlying horse state space $\mathcal{S}$. The set of risky events, denoted by $\mathcal{R}$, consists of all subintervals of $[0, 1]$, with each interval $R \in \mathcal{R}$ assigned a probability equal to its length. Equivalently, one may interpret risky events as being generated by a randomising device that draws a number uniformly from $[0, 1]$. That is, for any interval $R = [a, b] \subseteq [0, 1]$, the probability of R is given by $\lambda(R) = b - a$, where λ denotes the Lebesgue measure. We sometimes refer to $[0, 1]$ as the risky state space. Formally, $\mathcal{H}$ and $\mathcal{R}$ are taken to be sub σ -algebras of $\mathcal{E}$, containing all ambiguous and all risky events,

respectively.

Outcomes. The second as well as the final primitive of the model under uncertainty is the set of *outcomes*, also frequently called *consequences*:

$$\mathcal{X} = \{\dots, x, y, \dots\} \text{ is the set of } \textit{outcomes}.$$

An outcome x captures everything (relevant) about the DM's well-being. The set of outcomes can be finite or infinite. States and outcomes are the only primitives in our framework; all other definitions can be derived from them.

Acts. Having introduced the first and second primitives of the uncertain world, we now turn to acts, which are constructed from these two primitives. Formally, an *act* is a function from the set of states to the set of outcomes, $h : \mathcal{S} \rightarrow \mathcal{X}$. That is, for each state $s \in \mathcal{S}$, the act assigns an outcome $h(s) \in \mathcal{X}$. The outcome $h(s)$ is the outcome that the DM would receive if the state s actually obtains. For instance, acts may correspond to monetary bets on horses. Because the DM is uncertain about which horse will win, she is consequently uncertain about the monetary payoff resulting from a bet. For an act h and outcome x , $h^{-1}(x) = \{s : h(s) = x\} \in \mathcal{E}$ is a horse event. It is the event such that, if $s \in h^{-1}(x)$, then the outcome x is obtained. The set of acts is written as:

$$\mathcal{A} = \{\dots, h(\cdot), \dots\} \text{ is the set of } \textit{acts} \text{ on } \mathcal{S}.$$

Simple acts are acts that have a finite number of outcomes. The generic notation of a simple act is

$$h = (H_1, x_1; \dots; H_n, x_n).$$

This act yields outcome x_i if the true state belongs to event H_i . This notation implies that the events $(H_1, \dots, H_n)$ partition the state space $\mathcal{S}$. An act is *constant* if $h(s) = h(s') = x$ for all $s, s' \in \mathcal{S}$. A constant act always gives the same outcome, so we sometimes identify a constant act as $h = x$.

Gambles. Following Sarin and Wakker's approach, our notation introduced so far is summarised as follows.

$\mathcal{X}$	An arbitrary set of <i>outcomes</i> ;
$(H_1, x_1; \dots; H_n, x_n)$	An <i>act</i> yielding outcome x_i in event H_i ;
$\mathcal{A}$	The set of all acts;
$(R_1, x_1; \dots; R_n, x_n) = (p_1, x_1; \dots; p_n, x_n)$	A <i>lottery</i> yielding x_i in risky event R_i with probability p_i ($\sum_{i=1}^n p_i = 1$, $n < \infty$);
$\mathcal{L}$	The set of all lotteries;
$f = (A_1, x_1; \dots; A_n, x_n)$	A <i>gamble</i> , i.e. either an act or a lottery;
$\mathcal{G} = \mathcal{A} \cup \mathcal{L}$	The set of all gambles;
$\succsim$	The DM's preference relation on $\mathcal{G}$.

Acts and lotteries are generically referred to as *gambles*, the objects of choice throughout this paper. Therefore, the binary relation $\succsim$ denotes the DM's preference relation over $\mathcal{G}$. As usual, we write $f \succ g$ if $[f \succsim g \text{ and not } g \succsim f]$, and $f \sim g$ if $[f \succsim g \text{ and } g \succsim f]$. The relations $\leq$ and $<$ are defined analogously. Acts are functions from the horse state space $\mathcal{S}$ to $\mathcal{X}$, so that the ambiguous events $H_1, \dots, H_n$ partition the horse state space $\mathcal{S}$. Lotteries are functions from $[0, 1]$ to $\mathcal{X}$, and the risky events $R_1, \dots, R_n$ partition the risky state space $[0, 1]$. Outcomes are identified with degenerate lotteries as well as constant acts, hence we use the same preference relation $\succsim$ on $\mathcal{X}$ and write $x \succsim y$ for $x, y \in \mathcal{X}$. An event $A \in \mathcal{E}$ is *null* if $f_A h \sim g_A h$ for all $f, g, h \in \mathcal{G}$; otherwise, it is *non-null*. An event is null if its outcomes never affect preference. Loosely speaking, a null event is equally likely as the impossible event.

In addition, we simplify the notation for conditioned gambles by defining f_A as the restriction of gamble f to event A . We denote by $f_A h$ the gamble that takes the value $f(s)$ for all $s \in A$ and $h(s)$ for all $s \in \mathcal{S} \setminus A$. Alternatively, the gamble $f_A h$ can be expressed in matrix form as

$$f_A h = \begin{bmatrix} f(s) & \text{if } s \in A \\ h(s) & \text{if } s \notin A \end{bmatrix}.$$

We will use these expressions interchangeably throughout the paper.

2.2 Calibrating Probabilities of Ambiguous Events

This section outlines how Sarin and Wakker (1992, 1997) calibrate utilities and subjective probabilities through the use of a randomising device, such as a roulette wheel. The problem considered here is to evaluate an act $(H_1, x_1; \dots; H_n, x_n)$ that yields outcome x_i if event H_i occurs, where the probabilities of events H_i ($i = 1, \dots, n$) are not given. The collection $\{H_i\}_{i=1}^n$ is assumed to be disjoint and exhaustive. To calibrate a DM's beliefs, we introduce a randomising device, such as a roulette wheel, which generates physical (known or objective) probabilities. Using this device, any desired probability distribution over outcomes can be specified by a lottery contingent on the realisation of the roulette wheel.

Assumption 2.1 (Elicitation of Utilities). Preferences over lotteries $\mathcal{L}$ are represented by rank-dependent utility.

The rank-dependent utility value of a lottery $(p_1, x_1; \dots; p_n, x_n)$ with $x_1 < \dots < x_n$ is given by $u(x_1) + \sum_{i=2}^n w(p_i + \dots + p_n) \cdot [u(x_i) - u(x_{i-1})]$, where u is the utility function as in expected utility, and w is the probability weighting function, i.e. a strictly increasing function from $[0, 1]$ to $[0, 1]$, satisfying $w(0) = 0$ and $w(1) = 1$.

A way to calibrate the probability of an ambiguous event H is as follows. First, we fix a favourable outcome $\bar{x}$ and an unfavourable outcome $\underline{x}$. Then, a roulette-wheel probability α is elicited such that the DM is indifferent between the act $(H, \bar{x}; \text{not-}H, \underline{x})$ and the lottery $(\alpha, \bar{x}; 1 - \alpha, \underline{x})$. This indifference is displayed in Figure 1.

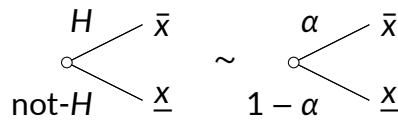


Figure 1: Elicitation of Probabilities

The method for eliciting probabilities just described is based on a comparison of likelihoods of different events through bets on events. Events E and E' are equally likely if the DM is indifferent to betting on E and betting on E' , i.e. $(E, \bar{x}; \text{not-}E, \underline{x})$ is indifferent to $(E', \bar{x}; \text{not-}E', \underline{x})$. In this case, we write $E \sim^L E'$. A formal definition of the likelihood ordering $\succsim^L$ over events is provided in Section 2.4. This method for calibrating probabilities has been well understood throughout history (see Borel, 1939, Sections 39 and 48).

Assumption 2.2 (Elicitation of Probabilities). For any horse event $H \in \mathcal{H}$, there exists a roulette event $R \in \mathcal{R}$ such that $H \sim^L R$.

Once the probabilities of ambiguous events have been calibrated, the value of an act $(H_1, x_1; \dots; H_n, x_n)$ can be elicited as follows. The DM first assigns probabilities $p_1, \dots, p_n$ to the events $H_1, \dots, H_n$ using a randomising device, such as a roulette wheel. A corresponding (*matching*) lottery $(R_1, x_1; \dots; R_n, x_n)$ is then constructed, where the risky events $R_1, \dots, R_n$ occur with probabilities $p_1, \dots, p_n$, respectively. For consistency, we want the value of the act $(H_1, x_1; \dots; H_n, x_n)$ to be the same. The following condition ensures this consistency.

Assumption 2.3 (Valuation of Acts). Assume that $x_1 < \dots < x_n$. An act $(H_1, x_1; \dots; H_n, x_n)$ is indifferent to a lottery $(R_1, x_1; \dots; R_n, x_n)$ whenever $\cup_{j=i}^n H_j \sim^L \cup_{j=i}^n R_j$ for all $i = 1, \dots, n$.

Assumption 2.3 is based on the principle that two gambles are judged indifferently if, for each outcome, the event of receiving the better outcome is equally likely under both gambles. This method of evaluating an uncertain act by matching it with a lottery that has the same upper cumulative probability distribution has been widely used in the decision theory literature; see Sarin and Wakker (1992, 1997).

2.3 Capacities and Choquet Expected Utility

A set function $\nu : \mathcal{E} \rightarrow [0, 1]$ is called a *capacity* if $\nu(\emptyset) = 0$, $\nu(\mathcal{S}) = 1$, and ν is *monotonic with respect to set inclusion*, i.e. $A \subset B \implies \nu(A) \leq \nu(B)$ for all $A, B \in \mathcal{E}$. A capacity reduces to a *probability measure* if it is *additive*, that is, $\nu(A \cup B) = \nu(A) + \nu(B)$ for all disjoint $A, B \in \mathcal{E}$. Thus, a probability measure is an additive capacity, whereas a capacity generalises the notion of probability by relaxing additivity to monotonicity. In this sense, capacities are often referred to as *non-additive probabilities*: they resemble probability measures but do not, in general, satisfy the additivity axiom. Further, a capacity $\nu : \mathcal{E} \rightarrow [0, 1]$ is said to be *convex-valued* if, for all $\alpha \in [0, 1]$, there exists $A \in \mathcal{E}$ such that $\nu(A) = \alpha$. A capacity $\nu : \mathcal{E} \rightarrow [0, 1]$ is a *probability transformation* if there is a strictly increasing function $\phi : [0, 1] \rightarrow [0, 1]$ and a probability measure $p : \mathcal{E} \rightarrow [0, 1]$ such that $\nu = \phi \circ p$.

For a capacity ν and a function $f : \mathcal{S} \rightarrow \mathbb{R}$, the *Choquet integral* of f with respect to ν ,

denoted $\int_S f \, dv$ or $\int f \, dv$ and introduced by Choquet (1954), is

$$\int_0^\infty v(\{s \in S : f(s) \geq x\}) \, dx + \int_{-\infty}^0 [v(\{s \in S : f(s) \geq x\}) - 1] \, dx. \quad (1)$$

Note that for non-negative f the second term on the right hand vanishes. The Choquet integral is a way to extend the Lebesgue integral to a special class of capacities. Wakker (1989, Chapter VI) gives more illustrations for the Choquet integral.

The DM exists only to maximise a preference relation $\succsim$ defined over $\mathcal{G}$. A function $V : \mathcal{G} \rightarrow \mathbb{R}$ represents $\succsim$ if $f \succsim g \iff V(f) \geq V(g)$. We say that *Choquet expected utility* (CEU) applies if there exist a *utility function* $u : \mathcal{X} \rightarrow \mathbb{R}$ and a capacity v on S such that $f \mapsto \int_S u \circ f \, dv$ represents $\succsim$. The integral is the CEU of f , denoted $CEU(v, u)(f)$. CEU is also known as *rank-dependent utility for unknown probabilities*. The CEU of a rank-ordered gamble $f = (A_1, x_1; \dots; A_n, x_n)$, with $x_1 < \dots < x_n$, is given by

$$CEU(v, u)(f) = u(x_1) + \sum_{i=2}^n v(A_i \cup \dots \cup A_n) \cdot [u(x_i) - u(x_{i-1})]. \quad (2)$$

This expression can be seen to equal the Choquet integral defined in (1), and this result will be useful in the sequel. It can be understood as building utility marginally from the worst outcome upward. The term $u(x_1)$ represents the utility of the worst possible outcome, which is received for sure in the sense that all states yield at least x_1 . CEU starts from this baseline. For each $i \geq 2$, the difference $u(x_i) - u(x_{i-1})$ measures the marginal utility gain that moves from outcome x_{i-1} to the better outcome x_i . However, this gain only matters in those states where outcomes of at least x_i are realised, captured by $A_i \cup \dots \cup A_n$, i.e. the event of receiving outcome at least as good as x_i . Instead of weighting marginal utility gain by cumulative probabilities in expected utility, CEU weights them by the capacity $v(A_i \cup \dots \cup A_n)$, which reflects the DM's perceived importance or plausibility of achieving outcomes at least as good as x_i .

If we expand all terms of (2) and rearrange them, the CEU evaluation of an act f can

alternatively be written as:

$$CEU(v, u)(f) = \sum_{i=1}^n \left[\underbrace{v(A_i \cup \dots \cup A_n) - v(A_{i+1} \cup \dots \cup A_n)}_{\text{The decision weight of event } A_i, \text{ denoted } \pi_i} \right] \cdot u(x_i),$$

where u is the utility function for outcomes and v is a capacity on the state space $\mathcal{S}$. The decision weight π_i depends on the ranking of outcome x_i , for $i = 1, \dots, n$; in particular, $\pi_n = v(A_n)$. More generally, π_i is given by the marginal v contribution of this event A_i to the *dominating event* $D_i = A_{i+1} \cup \dots \cup A_n$, which corresponds to receiving a strictly better outcome. Formally, the decision weight π_i can be rewritten as $\pi_i = v(A_i \cup D_i) - v(D_i) =: d(A_i, D_i)$. In this sense, CEU is grounded in the intuition of rank dependence: the decision weight π_i can be interpreted as the attention paid to an event, which depends not only on the event itself but also on the favourability of the outcome it yields in comparison to the outcomes yielded with other events.

The non-additive treatment of uncertainty is intended to express the DM's aversion of uncertainty. The more confidence a DM has in his subjective probabilities, the more these will be additive. Risk is an extreme case in which there is full confidence in the correctness of the probabilities. If the capacity v in the above expression is replaced by an additive capacity p , it immediately becomes subjective expected utility (Savage, 1954). Preferences $\succsim$ over gambles $\mathcal{G}$ conform to *subjective expected utility* (SEU) if the gamble $f = (A_1, x_1; \dots; A_n, x_n)$, with $x_1 < \dots < x_n$, is evaluated by

$$SEU(p, u)(f) = \int_{\mathcal{S}} u \circ f \, dp = \sum_{i=1}^n \left[p(A_i \cup \dots \cup A_n) - p(A_{i+1} \cup \dots \cup A_n) \right] \cdot u(x_i).$$

where $p : \mathcal{E} \rightarrow [0, 1]$ is a probability measure and $u : \mathcal{X} \rightarrow \mathbb{R}$ is a utility function for outcomes. Note that the integral used in SEU is in the sense of the Lebesgue integral. Alternatively, SEU can be expressed in terms of cumulative probabilities as follows:

$$SEU(p, u)(f) = u(x_1) + \sum_{i=2}^n p(A_i \cup \dots \cup A_n) \cdot [u(x_i) - u(x_{i-1})].$$

A capacity v is said to be *convex* if $v(A \cup B) \geq v(A) + v(B) - v(A \cap B)$ holds for any events $A, B \in \mathcal{E}$. The capacity v is *concave* if the reversed inequality holds. A CEU maximiser is

pessimistic if and only if the capacity is convex, and optimistic if and only if the capacity is concave. *Ambivalence* is a composition of pessimistic and optimistic responses to uncertainty, which is a prevalent attitude found in experiments. If capacities are initially concave and then convex, ambivalence is called *inverse-S behaviour*.

2.4 Revealed Likelihood Ordering of Events

Following Savage (1954), we define a *subjective likelihood order* $\succsim^L$ on events $\mathcal{E}$ from $\succsim$ on gambles $\mathcal{G}$, written $A \succsim^L B$ and read, ‘A is, subjectively, more likely than B,’ if there exist outcomes $x \succ y$ such that $x_A y \succ x_B y$. Definition 2.1 of Abdellaoui and Wakker (2005) is referred to as the same condition as *revealed more likely in a basic sense*. $\prec^L, \succsim^L, \leq^L$ and $\sim^L$ are defined in the usual way. It can be shown that $\succsim^L$ is a well-defined weak order over $\mathcal{E}$. To see this point, we give a detailed explanation for revealing likelihood orderings of events from gambles on events. Let A and B designate events, such as rain tomorrow or Dow Jones going up tomorrow. A^c and B^c denote complementary events. y designates an outcome, i.e. a constant gamble $(A, y; A^c, y) = (B, y; B^c, y)$. x (‘good’) denotes an outcome better than y (‘bad’), i.e. $x \succ y$. An agent, whose status quo is y , can choose between betting on A or on B. Betting on A means improving y to x under A, and it designates the gamble $(A, x; A^c, y)$. Betting on B means improving y to x under B, and it designates the gamble $(B, x; B^c, y)$.

Definition 1 (Abdellaoui and Wakker, 2005). Event A is *revealed more likely* than event B in a *basic sense* ($A \succ^L B$) if there exist outcomes $x \succ y$ such that the following configuration holds:

$$\begin{array}{cc|cc} A & A^c & & B & B^c \\ \hline y & y & = & y & y \\ x & y & \succ & x & y \end{array}$$

If such a configuration holds with $\sim$ (or $<$) instead of $\succ$, then we write $A \sim^L B$ (or $A \prec^L B$), and call A *revealed equally likely as* (revealed less likely than) B in the basic sense.

Under CEU, the constant gamble $(A, y; A^c, y)$ is evaluated by $u(y) = v(A)u(y) + [1 - v(A)]u(y)$, the constant gamble $(B, y; B^c, y)$ is evaluated by $u(y) = v(B)u(y) + [1 - v(B)]u(y)$, the gamble $(A, x; A^c, y)$ is evaluated by $v(A)u(x) + [1 - v(A)]u(y)$, and the gamble $(B, x; B^c, y)$ is evaluated

by $v(B)u(x) + [1 - v(B)]u(y)$. Therefore, betting on A entails an improvement $v(A)[u(x) - u(y)]$, and betting on B entails an improvement $v(B)[u(x) - u(y)]$. Revealed likelihood then measures capacity orderings:

$$A \succ^L B \iff v(A) > v(B);$$

$$A \sim^L B \iff v(A) = v(B);$$

$$A \prec^L B \iff v(A) < v(B).$$

Throughout this paper, the more-likely-than relation will correspond to a ‘capacity’, i.e. a non-additive probability.

3 Updating Choquet Beliefs

Formally, a preference relation $\succsim$ on $\mathcal{G}$ is said to have a CEU representation, denoted (u, v) , if for all gambles $f, g \in \mathcal{G}$,

$$f \succsim g \iff \text{CEU}(v, u)(f) \geq \text{CEU}(v, u)(g) \iff \int_{\mathcal{S}} u \circ f \, dv \geq \int_{\mathcal{S}} u \circ g \, dv,$$

where $u : \mathcal{X} \rightarrow \mathbb{R}$ is the utility function for outcomes and v is a capacity on $\mathcal{S}$.

Given a preference relation $\succsim$ on $\mathcal{G}$ that has a CEU representation, we have already known that $A \succsim^L B$ if and only if $v(A) \geq v(B)$. Consider a DM whose preference relation $\succsim$ on $\mathcal{G}$ incorporates beliefs that are represented by a finitely-additive subjective probability measure, p on $\mathcal{E}$, such that $A \succsim^L B$ if and only if $p(A) \geq p(B)$. Thus, $v(A) \geq v(B)$ if and only if $p(A) \geq p(B)$.

The DM is assumed to be characterised by a *class* of conditional preference relations on $\mathcal{G}$. For each event $E \in \mathcal{E}$, let $\succsim_E$ be a *conditional preference relation on $\mathcal{G}$ given E* . It represents the agent’s preferences after it becomes known that the non-null event E has occurred (or if she knew that E had obtained). $\preceq_E, \succ_E, \prec_E$ and $\sim_E$ are defined as usual. Let $\{\succsim_E\}_{E \in \mathcal{E}}$ be a collection of conditional preference relations. The relation $\succsim$ denotes the individual’s ex-ante *unconditional preference relation on $\mathcal{G}$* . If $E = \mathcal{S}$, $\succsim_{\mathcal{S}}$ becomes the unconditional preference relation on $\mathcal{G}$, simply denoted by $\succsim$. Given an event $E \in \mathcal{E}$, a *conditional* or

updated capacity v_E is a capacity conditional on E , i.e. for all A in $\mathcal{E}$ with $A \cap E = E$, $v_E(A) = 1$. When $E = \mathcal{S}$, $v_{\mathcal{S}}$ is considered as the *unconditional capacity*, simply written as v . Throughout this paper, it is assumed that every conditional preference relation $\succsim_E$ of $\{\succsim_E\}_{E \in \mathcal{E}}$ has the following representation:

Definition 2. Let $E \in \mathcal{E}$ be an event, v_E be a capacity, and $u : \mathcal{X} \rightarrow \mathbb{R}$ be a utility function. A conditional preference $\succsim_E$ is said to be *represented by a utility function u and a capacity v_E* , denoted (u, v_E) , if

$$f \succsim_E g \iff CEU(v_E, u)(f) \geq CEU(v_E, u)(g) \iff \int_{\mathcal{S}} u \circ f \, dv_E \geq \int_{\mathcal{S}} u \circ g \, dv_E.$$

If $E = \mathcal{S}$, $u_{\mathcal{S}}$ becomes u , that is, $\succsim$ is represented by (u, v) .

4 Existing Updating Rules for Probabilities and Capacities

In this section, we review the Bayesian updating rule for probabilities and three existing updating rules for capacities. Meanwhile, we present the notions of dynamic consistency and consequentialism.

4.1 The Bayesian Updating Rule for Probabilities

To investigate the effects of obtaining new information on individual decision making, we consider the following updating rules. In probability theory, it is common to consider the Bayes' rule as the unique updating rule. Let p be a probability measure on a measurable space $(\mathcal{S}, \mathcal{E})$ and let $p(E) > 0$ for $E \in \mathcal{E}$. Observing such an event A , the *Bayes' rule* is defined by

$$p_E(A) = \frac{p(A \cap E)}{p(E)} \quad \text{for all } A \text{ in } E. \quad (3)$$

For preferences which are represented by a CEU functional with an additive capacity v , the standard Bayesian updating rule satisfies the following three axioms (see Ghirardato, 2002).

Axiom SI (State Independence). For any two outcomes $x, y \in \mathcal{X}$ and any non-null event

$E \in \mathcal{E}$,

$$x \succcurlyeq y \iff x \succcurlyeq_E y.$$

State independence requires the conditional preferences over outcomes to agree with the unconditional preferences over outcomes.

Axiom C (Consequentialism). For any two acts $f, g \in \mathcal{A}$ and any event $E \in \mathcal{E}$,

$$f = g \text{ on } E \implies f \sim_E g.$$

Consequentialism rules out the effects on future choices from the outcomes which would have become relevant in the event E^c , which did not happen. The DM cares about consequences, but not how consequences are achieved.

Axiom DC (Dynamic Consistency). For any acts $f, g \in \mathcal{A}$ and any non-null event $E \in \mathcal{E}$,

$$f \succ_E g \iff f_E g \succ g.$$

Dynamic consistency links conditional and unconditional preferences by requiring the preferences after E occurred remain consistent with ex-ante preferences. Dynamic consistency means that, as long as the DM has followed the ex-ante preferred plan so far, she never wishes to deviate from the plan.

Example 1 (Ellsberg, 1961). Suppose that an urn contains 90 balls. The number of red balls is 30, and the number of black and yellow balls is 60. Consider a bet on drawing a ball from this urn. The DM will get \$0 or \$100 depending on the colour. The DM has to choose

between two pairs of bets f vs. f' and g vs. g' :

$$f = \begin{bmatrix} \$100 & \text{if } s \in R \\ \$0 & \text{if } s \in B \\ \$0 & \text{if } s \in Y \end{bmatrix} \quad f' = \begin{bmatrix} \$0 & \text{if } s \in R \\ \$100 & \text{if } s \in B \\ \$0 & \text{if } s \in Y \end{bmatrix}$$

$$g = \begin{bmatrix} \$100 & \text{if } s \in R \\ \$0 & \text{if } s \in B \\ \$100 & \text{if } s \in Y \end{bmatrix} \quad g' = \begin{bmatrix} \$0 & \text{if } s \in R \\ \$100 & \text{if } s \in B \\ \$100 & \text{if } s \in Y \end{bmatrix}.$$

For example, bet f' states that we obtain \$100 if we draw a black ball and nothing otherwise. Many respondents prefer f to f' and g' to g , denoted by $f \succ f'$ and $g' \succ g$, respectively. However, these choices cannot be explained within the framework of SEU, which is called *Ellsberg's paradox*.

Now we reformulate this static example in a dynamic version, which is based on Dominiak and Lefort (2011).

Example 2 (Dominiak and Lefort, 2011). We consider two stages, the ex-ante stage and the interim stage. The ex-ante stage ($t = 0$) corresponds to Example 1 in which the DM has no information. Then, many respondents prefer f to f' and g' to g , denoted by $f \succ f'$ and $g' \succ g$, respectively. In the interim stage ($t = 1$), one ball is drawn from the urn at random and the DM is informed that the ball is not yellow, i.e. $\{R, B\}$. We denote such a conditional preference by $\succ_{\{R, B\}}$. On the one hand, since $f \succ f'$ and $f = f'$ on $\{R, B\}^c$, dynamic consistency implies $f \succ_{\{R, B\}} f'$. Similarly, since $g' \succ g$ and $g' = g$ on $\{R, B\}^c$, dynamic consistency implies $g' \succ_{\{R, B\}} g$. On the other hand, since $f = g$ on $\{R, B\}$, consequentialism implies $f \sim_{\{R, B\}} g$. Similarly, since $f' = g'$ on $\{R, B\}$, consequentialism implies $f' \sim_{\{R, B\}} g'$. Furthermore, consequentialism implies that if $f \succ_{\{R, B\}} f'$, then $g \succ_{\{R, B\}} g'$ and vice versa. Therefore, if conditional preferences retain dynamic consistency, then consequentialism is violated. On the contrary, if conditional preferences retain consequentialism, dynamic consistency is violated.

Dominiak and Lefort (2011) point out that when considering updating rules under ambiguity, dynamic consistency and consequentialism cannot hold simultaneously, so that one

of the properties must be dropped. Dominiak et al. (2012) experimentally find that a significant majority subjects violate dynamic consistency rather than consequentialism. For this reason, we will retain consequentialism and drop dynamic consistency in this paper.

4.2 Three Existing Updating Rules for Capacities

The formal definitions of the three updating rules for capacities introduced in Section 1 are as follows.

Definition 3. The *naïve Bayes' updating rule* or *optimistic updating rule* is defined by

$$v_E^O(A) = \frac{v(A \cap E)}{v(E)}, \quad (4)$$

the *Dempster-Shafer updating rule* or *pessimistic updating rule* is defined by

$$v_E^{DS}(A) = \frac{v(A \cup E^c) - v(E^c)}{1 - v(E^c)}, \quad (5)$$

and the *generalised/full Bayesian updating rule* is defined by

$$v_E^{GB}(A) = \frac{v(A \cap E)}{v(A \cap E) + 1 - v(A \cup E^c)}, \quad (6)$$

where for any non-null event $E \in \mathcal{E}$ which is likely to occur, E^c denotes the complement of E and all the denominators are assumed to be non-zero.

If v is additive, then all three updating rules for capacities coincide with the standard Bayesian updating rule for probabilities in Eq. (3). Suppose that v reduces to an additive probability measure p . The result for the optimistic updating rule in Eq. (4) follows immediately. For the pessimistic updating rule in Eq. (5), additivity implies $1 - p(E^c) = p(E)$. Moreover, $A \cup E^c = (A \cap E) \cup E^c$, where the union is disjoint. Hence, $p(A \cup E^c) = p(A \cap E) + p(E^c)$, and therefore $p(A \cup E^c) - p(E^c) = p(A \cap E)$. Finally, for the full Bayesian updating rule in Eq. (6), additivity yields $1 - p(A \cup E^c) = p((A \cup E^c)^c)$. By De Morgan's law, $(A \cup E^c)^c = A^c \cap (E^c)^c = A^c \cap E$, so that $1 - p(A \cup E^c) = p(A^c \cap E)$. Applying additivity once more, since $E = (A \cap E) \cup (A^c \cap E)$ is a disjoint union, we obtain $p(A \cap E) + p(A^c \cap E) = p(E)$.

When CEU preferences exhibit probabilistic sophistication, the capacity admits a representation of the form $\nu = \phi \circ p$, where the additive probability measure p represents beliefs and the transformation function ϕ captures attitudes towards uncertainty. This will be discussed in detail in Section 6.3. More generally, Wakker (2001) shows how a capacity ν exhibits the DM's uncertainty attitude. A natural question is whether these updating rules are compatible with probabilistic sophistication. Note that preserving probabilistic sophistication under updating requires that beliefs be updated via Bayes' rule at the level of p , yielding a posterior p_E , while the transformation ϕ remains unchanged, so that the updated capacity takes the form $\nu_E = \phi \circ p_E$. However, existing updating rules for capacities update the capacity directly and generally fail to preserve this structure, implicitly entangling belief revision with changes in uncertainty attitudes, i.e. $\nu_E = \phi_E \circ p_E$. This motivates the need for an alternative updating procedure that maintains a clean separation between beliefs and attitudes towards uncertainty, with belief updating proceeding in a Bayesian manner.

5 Dynamic Preference Foundations for $\nu_E = \phi \circ p_E$

The first axiom requires each conditional preference $\succsim_E$ to be a *weak order* on $\mathcal{G}$. As usual, we use $\succ_E$ (resp. $\sim_E$) to denote the asymmetric (resp. symmetric) component of a weak order $\succsim_E$.

Axiom 1 (Weak Ordering). For each event $E \in \mathcal{E}$, $\succsim_E$ is a complete and transitive binary relation on $\mathcal{G}$.

In the literature, the following axiom is often referred to as *consequentialism*. However, the idea captured here does not fully coincide with the original meaning of consequentialism (see Hammond, 1988). For this reason, we follow Arrow's terminology and refer to it instead as *conditional preference*, in the sense of Arrow (1966). Conditional preference rules out the effects on future choices from outcomes that would have been realised had the event E^c occurred, since E^c did not occur. More specifically, conditional preference requires that preferences conditional on a non-null event E are not affected by outcomes outside the conditional event E^c . Intuitively, once the DM is informed that an event E has occurred, only the uncertainty about all subevents of E is relevant for her preferences, and the uncertainty

about counterfactual events in E^c no longer affects future choices.

The conditional preferences are ‘forward-looking’ in the sense that what happens off the conditioning event should not be able affect the conditional preference between any pairs of gambles. For a given conditional preference relation $\succsim_E$, we do not need explicitly to keep track of what outcomes might have resulted from a gamble that had a state outside of E obtained.

Axiom 2 (Conditional Preference). For any non-null event $E \in \mathcal{E}$ and all gambles $f, g \in \mathcal{G}$,

$$f(s) = g(s) \text{ for each } s \in E \implies f \sim_E g.$$

Proposition 1. Assume Axiom 1 (Weak Ordering) and Axiom 2 (Conditional Preference). Then for all events A, B and E such that $A \cup B \subseteq E$, any gamble $g \in \mathcal{G}$ and outcomes x and y with $x \succ y$,

$$A \succsim_E^L B \iff \begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \in E \setminus A \\ g(s) & \text{if } s \notin E \end{bmatrix} \succsim_E \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \in E \setminus B \\ g(s) & \text{if } s \notin E \end{bmatrix}.$$

Dynamic consistency links conditional and unconditional preferences by requiring that ex-post preferences after some likely event E occurred remain consistent with ex-ante preferences. Two-outcome dynamic consistency requires that if the outcome space has only two outcomes x and y with $x \succ y$, then no matter what event would occur, the ranking of x and y would not change.

Axiom 3 (Two-Outcome Dynamic Consistency). For any non-null event $E \in \mathcal{E}$, all events $A, B \subseteq E$, and all outcomes $x \succ y$,

$$\begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \notin A \end{bmatrix} \succsim \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \notin B \end{bmatrix} \iff \begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \notin A \end{bmatrix} \succsim_E \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \notin B \end{bmatrix}$$

Sarin and Wakker (2000) restrict Savage’s sure-thing principle P2 to two-outcome gambles P2*. To see that P2* is a restriction of P2, we observe that in P2* a common outcome y under event C in the first preference is replaced by a common outcome x in the second preference.

Axiom P2* (*Two-Outcome Sure-Thing Principle*; *Sarin and Wakker, 2000*). For all outcomes $x \succ y$ and events A, B, C with $A \cap C = B \cap C = \emptyset$,

$$\begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \notin A \end{bmatrix} \succcurlyeq \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \notin B \end{bmatrix} \iff \begin{bmatrix} x & \text{if } s \in A \cup C \\ y & \text{if } s \notin A \cup C \end{bmatrix} \succcurlyeq \begin{bmatrix} x & \text{if } s \in B \cup C \\ y & \text{if } s \notin B \cup C \end{bmatrix}.$$

Proposition 2. *Assume Axiom 1 (Weak Ordering). Then Axiom 2 (Conditional Preference) and Axiom 3 (Two-Outcome Dynamic Consistency) jointly imply Axiom P2* (Two-Outcome Sure-Thing Principle).*

Now we present the second key axiom of this paper. Consider the gamble $f = (f^{-1}(x_0), x_0; \dots; f^{-1}(x_n), x_n)$ where outcomes are enumerated from worst to best, $x_0 < \dots < x_n$. For $f \in \mathcal{G}$, the *decumulative representation* F is written as follows:

$$F = (F_1, \dots, F_n) \quad \text{where } F_i = \bigcup_{j=i}^n f^{-1}(x_j).$$

Each F_i is the union of all events for which the gamble f yields x_i or any other preferred outcome. Clearly, $F_n = f^{-1}(x_n)$ for all $f \in \mathcal{G}$ as no outcome is preferred to x_n . Also, one would find $F_0 = \mathcal{S}$ for all gambles. Therefore, we drop the F_0 from the notation and write $F = (F_1, \dots, F_n)$. No relevant information is lost when transforming a gamble to its decumulative representation, that is, $F = G$ if and only if $f = g$. Given the rank ordering of outcomes, the CEU of a decumulative gamble is given by

$$CEU(F) = u(x_0) + \sum_{i=1}^n v(F_i) \left(u(x_i) - u(x_{i-1}) \right). \quad (7)$$

Now suppose that we observe the following two indifferences:

$$(A, F_2, \dots, F_n) \sim (B, G_2, \dots, G_n);$$

$$(A', F_2, \dots, F_n) \sim (B', G_2, \dots, G_n).$$

Substituting the CEU evaluation function in (7), the first indifference gives:

$$\begin{aligned} & u(x_0) + v(A) \left(u(x_1) - u(x_0) \right) + \sum_{i=2}^n v(F_i) \left(u(x_i) - u(x_{i-1}) \right) \\ &= u(x_0) + v(B) \left(u(x_1) - u(x_0) \right) + \sum_{i=2}^n v(G_i) \left(u(x_i) - u(x_{i-1}) \right); \end{aligned}$$

and the second indifference gives:

$$\begin{aligned} & u(x_0) + v(A') \left(u(x_1) - u(x_0) \right) + \sum_{i=2}^n v(F_i) \left(u(x_i) - u(x_{i-1}) \right) \\ &= u(x_0) + v(B') \left(u(x_1) - u(x_0) \right) + \sum_{i=2}^n v(G_i) \left(u(x_i) - u(x_{i-1}) \right). \end{aligned}$$

Deducting the second from the first and cancelling common terms gives:

$$v(A) - v(A') = v(B) - v(B').$$

Hence, two indifferences jointly reveal information about the capacity v differences.

Now let us define $F_i G := (G_1, \dots, G_{i-1}, F, G_{i+1}, \dots, G_n)$ in the usual way. Note that for $F_i G$ to be well-defined we must have $G_{i-1} \supset F \supset G_{i+1}$. We can write the above indifferences as $A_1 F \sim B_1 G$ and $A'_1 F \sim B'_1 G$. The role of a likelihood consistency axiom is to ensure that this revelation of differences does not depend on the ‘coordinate’ (here we use x_1 , but could be any x_i , where $i = 1, \dots, n$) or the other common parts. For example, under CEU, if the above indifferences hold and we also observe $A_i F \sim B_i G$, then we must have $A'_i F \sim B'_i G$. Again, notice that $A_i F$ must be well-defined, where rank dependence plays a role. This is, essentially, Gilboa’s P2* axiom (1987), or likelihood consistency of Abdellaoui and Wakker (2005).

Axiom RDLC (Rank-Dependent Likelihood Consistency). For all events $A, B, A', B' \in \mathcal{E}$ and

any coordinates $i, j = 1, \dots, n$, we have

$$\left[\begin{array}{l} A_i F \sim B_i G \quad \& \quad A_j Y \sim B_j Z \quad \& \\ A'_i F \sim B'_i G \end{array} \right] \Rightarrow A'_j Y \sim B'_j Z.$$

By Lemma A.1 of Sarin and Wakker (1992), we know that, for each event A , there exists a risky event $R \in \mathcal{R}$ such that $x_A y \sim x_R y$. This means that we can *match* subjective acts with lotteries. Let $\bar{x}$ be the best outcome and $\underline{x}$ be the worst outcome. $\bar{x}_\alpha \underline{x} := (\alpha, \bar{x}; 1 - \alpha, \underline{x})$ denotes the lottery in which the best outcome $\bar{x}$ is realised with probability $\alpha \in [0, 1]$ and the worst outcome $\underline{x}$ is realised with probability $1 - \alpha$. Then, we can match subjective acts with lotteries for all $A \in \mathcal{E}$ such that $\bar{x}_A \underline{x} \sim \bar{x}_\alpha \underline{x}$, get some new information E and find a new event $\tilde{A} \in \mathcal{E}$ such that $\bar{x}_{\tilde{A}} \underline{x} \sim_E \bar{x}_\alpha \underline{x}$. Graphically,

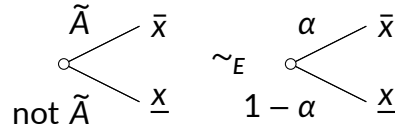


Figure 2: Elicitation of Updated Probabilities

More specifically, $\bar{x}_\alpha \underline{x}$ is the *ex-ante* indifference of an act $\bar{x}_A \underline{x} \sim \bar{x}_\alpha \underline{x}$ as well as the *ex-post* indifference of an act $\bar{x}_{\tilde{A}} \underline{x} \sim_E \bar{x}_\alpha \underline{x}$ after it becomes known that the non-null event E has occurred. If two indifferences $\bar{x}_A \underline{x} \sim \bar{x}_\alpha \underline{x}$ and $\bar{x}_{\tilde{A}} \underline{x} \sim_E \bar{x}_\alpha \underline{x}$ hold, then we write $A \sim^* \tilde{A}|E$, where $\sim^*$ is a new ‘equally likely’ relation based on matching.

Axiom 4 (Conditional Rank-Dependent Likelihood Consistency). For any non-null event $E \in \mathcal{E}$, all events $A, B, A', B', \tilde{A}, \tilde{B}, \tilde{A}', \tilde{B}' \in \mathcal{E}$ such that $A \sim^* \tilde{A}|E$, $A' \sim^* \tilde{A}'|E$, $B \sim^* \tilde{B}|E$ and $B' \sim^* \tilde{B}'|E$, and any coordinates $i, j = 1, \dots, n$, the following must hold:

$$\left[\begin{array}{l} A_i F \sim B_i G \quad \& \quad \tilde{A}_j Y \sim_E \tilde{B}_j Z \quad \& \\ A'_i F \sim B'_i G \end{array} \right] \Rightarrow \tilde{A}'_j Y \sim_E \tilde{B}'_j Z.$$

Axiom 4 ensures that the DM’s attitude towards uncertainty remains unchanged as her

beliefs are updated. To see this point, let us assume that CEU holds. The first pair of indifferences $A_i F \sim B_i G$ and $A'_i F \sim B'_i G$ imply that $v(A) - v(A') = v(B) - v(B')$ if and only if $\phi(p(A)) - \phi(p(A')) = \phi(p(B)) - \phi(p(B'))$. The second pair of indifferences $\tilde{A}_j Y \sim_E \tilde{B}_j Z$ and $\tilde{A}'_j Y \sim_E \tilde{B}'_j Z$ imply that $v_E(\tilde{A}) - v_E(\tilde{A}') = v_E(\tilde{B}) - v_E(\tilde{B}')$ if and only if $\phi_E(p_E(\tilde{A})) - \phi_E(p_E(\tilde{A}')) = \phi_E(p_E(\tilde{B})) - \phi_E(p_E(\tilde{B}'))$. By the definition of $\sim^*$ -relation introduced above, the DM reveals $p(A) = p_E(\tilde{A})$ and similarly $p(A') = p_E(\tilde{A}')$, $p(B) = p_E(\tilde{B})$ and $p(B') = p_E(\tilde{B}')$. Therefore, ϕ and ϕ_E must be positive affine transformations. Since $\phi(0) = 0$ and $\phi(1) = 1$, it must be $\phi = \phi_E$. For a detailed discussion, see the proof in the appendix following this paper.

Theorem 1. *Suppose that a DM's conditional preferences $\{\succsim_E\}_{E \in \mathcal{E}}$ are represented by CEU. Then the following statements are equivalent:*

- (i) *The class $\{\succsim_E\}_{E \in \mathcal{E}}$ satisfies Axioms 2–4: Conditional Preference, Two-Outcome Dynamic Consistency and Conditional Rank-Dependent Likelihood Consistency;*
- (ii) *There is a probability measure $p : \mathcal{E} \rightarrow [0, 1]$ such that the capacity $v : \mathcal{E} \rightarrow [0, 1]$ is represented by $v = \phi \circ p$ with a strictly increasing function $\phi : [0, 1] \rightarrow [0, 1]$. For each non-null event $E \in \mathcal{E}$, v is uniquely replaced by an updated capacity v_E associated with $\succsim_E$ such that $v_E = \phi \circ p_E$ and $p_E(A) = \frac{p(A \cap E)}{p(E)}$ for every non-null $E \in \mathcal{E}$.*

In Axiom 4, if $E = \mathcal{S}$, then A and $\tilde{A}$ are equally likely (and similarly $A' \sim^L \tilde{A}'$, $B \sim^L \tilde{B}$, and $B' \sim^L \tilde{B}'$). Hence, Axiom 4 reduces to Axiom RDLC whenever $E = \mathcal{S}$. Together with an appropriate Archimedean (continuity) axiom, this implies a CEU representation. Hence, CEU need not be assumed a priori, since Axiom 4 reduces to Axiom RDLC whenever $E = \mathcal{S}$, from which CEU follows under our proposed conditions, together with some additional technical assumptions (e.g. Monotone Continuity) and the solvability of each $\succsim_E$.

Remark. Suppose that a DM's conditional preferences are represented by a class of conditional preferences $\{\succsim_E\}_{E \in \mathcal{E}}$. Then the following are equivalent:

- (i) The class $\{\succsim_E\}_{E \in \mathcal{E}}$ satisfies Weak Ordering, Conditional Preference, Two-Outcome Dynamic Consistency, Conditional Rank-Dependent Likelihood Consistency and Monotone Continuity of Likelihood Orders;
- (ii) There is a utility function for outcomes $u : \mathcal{X} \rightarrow \mathbb{R}$, unique up to a positive affine trans-

formation, and a unique convex-ranged capacity $\nu : \mathcal{E} \rightarrow [0, 1]$ such that for all gambles $f, g \in \mathcal{G}$, the unconditional preference $\succsim$ is represented as the following:

$$f \succsim g \iff \int_{\mathcal{S}} u \circ f \, d\nu \geq \int_{\mathcal{S}} u \circ g \, d\nu,$$

where $\nu = \phi \circ p$ with a strictly increasing function $\phi : [0, 1] \rightarrow [0, 1]$ and a probability measure $p : \mathcal{E} \rightarrow [0, 1]$. Moreover, for each non-null event $E \in \mathcal{E}$ the conditional preference $\succsim_E$ is represented as the following:

$$f \succsim_E g \iff \int_{\mathcal{S}} u \circ f \, d\nu_E \geq \int_{\mathcal{S}} u \circ g \, d\nu_E,$$

with ν uniquely replaced by an updated capacity associated with the conditional preference $\succsim_E$, $\nu_E : \mathcal{E} \rightarrow [0, 1]$, where $\nu_E = \phi \circ p_E$ and $p_E(A) = \frac{p(A \cap E)}{p(E)}$ for each non-null event $E \in \mathcal{E}$.

Monotone Continuity was originally introduced by Villegas (1964) for preferences over events. Arrow (1970) extends this condition to a framework with acts, requiring the underlying collection of events $\mathcal{E}$ to be a σ -algebra. Kopylov (2010) further strengthens Arrow's version of Monotone Continuity by imposing it on nested sequences of events whose intersection is either empty—as in Villegas (1964) and Arrow (1970)—or a singleton.

6 Discussion

6.1 Comparison with Existing Choquet Updating Rules

The updating rule proposed in this paper differs from existing Choquet updating rules in both the object being updated and the order in which conditioning and probability distortion are applied. Suppose that the prior capacity is represented as

$$\nu = \phi \circ p,$$

where p is an additive probability measure and ϕ is a strictly increasing distortion function.

Existing CEU updating rules treat the capacity ν as the primitive object and update it directly when new information is observed. Thus, when $\nu = \phi \circ p$, they update an object in which probabilistic beliefs and probability distortion have already been combined. The optimistic, Dempster-Shafer, and generalised Bayesian rules are respectively given by

$$\begin{aligned} \nu_E^O(A) &= \frac{\nu(A \cap E)}{\nu(E)} = \frac{\phi(p(A \cap E))}{\phi(p(E))}, \\ \nu_E^{DS}(A) &= \frac{\nu(A \cup E^c) - \nu(E^c)}{1 - \nu(E^c)} = \frac{\phi(p(A \cup E^c)) - \phi(p(E^c))}{1 - \phi(p(E^c))}, \\ \nu_E^{GB}(A) &= \frac{\nu(A \cap E)}{\nu(A \cap E) + 1 - \nu(A \cup E^c)} = \frac{\phi(p(A \cap E))}{\phi(p(A \cap E)) + 1 - \phi(p(A \cup E^c))}. \end{aligned}$$

In the distorted probability representation, these rules can therefore be viewed as following a ‘distort-then-update’ approach.

Unlike the existing updating rules, the rule developed in this paper treats p and ϕ as conceptually distinct components of the capacity. The probability measure p represents the DM’s beliefs about the states of the world and is therefore the component that responds to new evidence. The distortion function ϕ represents a relatively stable behavioural attitude towards uncertainty. The proposed rule first updates p by Bayesian updating,

$$p_E(A) = \frac{p(A \cap E)}{p(E)},$$

and then applies the original distortion function:

$$\nu_E(A) = \phi(p_E(A)) = \phi\left(\frac{p(A \cap E)}{p(E)}\right).$$

It may therefore be described as an ‘update-then-distort’ procedure.

The distinction is substantive because reversing the order of updating and distortion does not generally yield the same result. For example, the optimistic rule yields

$$\nu_E^O(A) = \frac{\phi(p(A \cap E))}{\phi(p(E))},$$

whereas the proposed rule yields

$$v_E(A) = \phi \left(\frac{p(A \cap E)}{p(E)} \right).$$

For a general non-linear distortion function,

$$\frac{\phi(p(A \cap E))}{\phi(p(E))} \neq \phi \left(\frac{p(A \cap E)}{p(E)} \right).$$

Consequently, the two rules may generate different posterior capacities even when they start from the same prior probability and the same distortion function.

A central feature of our proposed rule is that the distortion function is preserved after information is received:

$$v = \phi \circ p \quad \implies \quad v_E = \phi \circ p_E.$$

Information changes the DM's probabilistic beliefs from p to p_E , while ϕ remains fixed. The rule therefore separates belief revision from the DM's attitude towards uncertainty. By contrast, direct updating of the capacity may induce a posterior capacity in which both the underlying beliefs and the distortion of those beliefs have changed.

The behavioural intuition can be illustrated by an investor whose decision depends on whether the economy enters a recession. Suppose that her assessed probability of a recession increases from $p(\text{Recession}) = 0.3$ to $p_E(\text{Recession}) = 0.5$ after she receives new information. Under the proposed rule, the capacity changes from $v = \phi \circ p$ to $v_E = \phi \circ p_E$. The investor becomes more concerned about a recession because she now thinks it as more likely, not because her attitude towards uncertainty has changed.

Similarly, suppose that a weather forecast increases a farmer's assessed probability of drought from $p(\text{Drought}) = 0.2$ to $p_E(\text{Drought}) = 0.4$. The forecast changes the farmer's belief about the likelihood of drought, while leaving unchanged the distortion function through which that belief is evaluated. It would be behaviourally implausible to interpret receiving a weather report as transforming the farmer into a fundamentally different type of DM. Rather, the evidence changes what he believes about the state of the world.

All of the updating rules considered above reduce to ordinary Bayesian updating when ϕ is the identity function, in which case the capacity is additive. The proposed rule may also coincide with an existing rule for particular classes of distortion functions. For example, it coincides with the optimistic rule when

$$\phi(x) = x^\gamma, \quad \gamma > 0,$$

because

$$\frac{\phi(p(A \cap E))}{\phi(p(E))} = \left(\frac{p(A \cap E)}{p(E)} \right)^\gamma = \phi\left(\frac{p(A \cap E)}{p(E)} \right).$$

For a general non-linear distortion function, however, the rules need not coincide. The distinguishing feature of the proposed rule is therefore that it updates the belief component while preserving the original distortion function. It consequently interprets learning as a revision of probabilistic beliefs rather than as a simultaneous revision of beliefs and uncertainty attitude.

One potential limitation of our proposed rule is that it relies on Bayesian updating,

$$p_E(A) = \frac{p(A \cap E)}{p(E)},$$

and therefore requires $p(E) > 0$. If the conditioning event E is assigned probability zero, the Bayesian posterior p_E cannot be constructed, and hence the updated capacity generated by applying the distortion function to p_E is also undefined. This limitation arises from the sequential structure of our updating rule, which first updates the additive probability measure and then applies the distortion. By comparison, the three existing updating rules for Choquet beliefs operate directly on the capacity and may therefore remain well defined even when the conditioning event has zero probability under the underlying probability measure (see Eichberger et al., 2010, Remark 1).

6.2 Probabilistic Sophistication

Essentially, this paper characterises a simple updating rule for probabilistic sophistication in the context of CEU. That is, an agent may quantify uncertainty using a prior probability

measure and update this prior to a posterior in a Bayesian manner, while maximising CEU with respect to her probabilistic beliefs. In a general setting, a probabilistically sophisticated agent needs not conform to expected utility, yet continues to express her beliefs by means of subjective probabilities. We call an agent *probabilistically sophisticated* if there exists a probability measure p over $\mathcal{S}$ such that:

- (i) the agent chooses between acts based on the probability distributions generated over the outcomes;
- (ii) first-order stochastic dominance holds if $f \succcurlyeq g$ whenever $p(\{s \in \mathcal{S} : f(s) \succcurlyeq x\}) \geq p(\{s \in \mathcal{S} : g(s) \succcurlyeq x\})$ for all outcomes x .

Such probabilistic sophistication is axiomatised by Machina and Schmeidler (1992), who drop Savage's axiom P2 (Sure-Thing Principle) and strengthen Savage's axiom P4 (Weak Comparative Probabilities). Further, Epstein and Le Breton (1993) generalise the results of Machina and Schmeidler (1992) and Grant (1995) by deriving probabilistic sophistication from dynamic consistency together with a set of additional assumptions. In their framework, all of Savage's axioms are maintained except for the sure-thing principle, and the assumption of consequentialism is not imposed. Such probabilistically sophisticated non-expected utility preferences are therefore able to accommodate the Allais paradox, which is well known to violate Savage's axiom P2.

Sarin and Wakker (2000) employ cumulative dominance—shown to be a necessary condition for CEU in their earlier work (Sarin and Wakker, 1992)—in conjunction with a weakening of Savage's axiom P2, which coincides with the two-outcome sure-thing principle (P2*) used in this paper, to obtain an alternative characterisation of probabilistically sophisticated beliefs. Formally, *cumulative dominance* holds if $f \succcurlyeq g$ whenever $\{s \in \mathcal{S} : f(s) \succcurlyeq x\} \succcurlyeq^L \{s \in \mathcal{S} : g(s) \succcurlyeq x\}$ for all outcomes x , where the preference between f and g is strict whenever one of the antecedent $\succcurlyeq^L$ orderings is strict ($>^L$). This condition says that if the cumulative event of receiving x or more is considered more likely under f than under g for every outcome x , then f should be preferred to g , where the more-likely-than relation $\succcurlyeq^L$ is derived from choices. Unlike stochastic dominance, the definition of cumulative dominance does not involve probabilities attached to events.

Grant et al. (2008) show that the conjunction of a weaker version of Savage's axiom P2, also termed the *two-outcome sure-thing principle* and denoted by P2', and Savage's original weak comparative probability axiom (P4) is equivalent to the strong comparative probability axiom proposed by Machina and Schmeidler (1992). Consequently, imposing Savage's original axioms while weakening P2 to the two-outcome sure-thing principle guarantees probabilistic sophistication. Moreover, Grant et al. (2008) further show that the conjunction of Sarin and Wakker's P2*, which coincides with ours and is a weaker version of P2', and the cumulative dominance axiom implies the conjunction of their axioms P2' and P4, whereas the reverse implication does not hold.

6.3 Uncertainty Attitudes

The capacity ν is *convex* if $\nu(C) + \nu(D) \leq \nu(C \cup D) + \nu(C \cap D)$ for all events $C, D \in \mathcal{E}$. ν is *concave* if the reversed inequality holds. Convexity can be rewritten as $\nu(D) - \nu(C \cap D) \leq \nu(C \cup D) - \nu(C)$. Equivalently, by substitution of $H = C \cap D$, $A = D \setminus C$, $I = C \setminus D$, convexity means that for all disjoint events A, H and I ,

$$\nu(A \cup H) - \nu(H) \leq \nu(A \cup H \cup I) - \nu(H \cup I).$$

It is equivalent to $d(A, H) \leq d(A, H \cup I)$, i.e.

$$d(A, H) \text{ is non-decreasing in } H.$$

Convexity of the capacity means that an event is assigned more decision weight when its associated outcome is ranked lower. Thus, convexity reflects a pessimistic attitude where more attention is paid to unfavourable outcomes than to favourable ones. Similar observations apply to concavity, with the inequalities reversed and an interpretation in terms of optimism.

Most studies assume convexity because it facilitates theoretical analysis and derivations. However, convexity implies that unlikely events associated with the highest outcomes are underweighted. Experimental evidence, by contrast, suggests in the opposite direction:

such events tend to be overweighted. This empirical pattern is commonly described as an *inverse-S shaped* capacity (Tversky and Fox, 1995; Fox et al., 1996; Fox and Tversky, 1998; Viscusi and Chesson, 1999; Wu and Gonzalez, 1999; Kilka and Weber, 2001). Inverse-S means that relatively high decision weights are assigned to unlikely events, while relatively low decision weights are assigned to likely events. In other words, a DM with an inverse-S shaped capacity tends to overestimate the likelihood of unlikely events and underestimate that of likely events.

CEU with probabilistic sophistication separates subjective beliefs from attitudes towards uncertainty. A DM's beliefs are represented by an additive probability measure, while her attitude towards uncertainty is captured by a probability transformation function ϕ . The function ϕ describes how the DM structures the realisation of the outcomes she cares about. In particular, it determines how beliefs are transformed into decision weights that reflect her attitude towards uncertainty. If ϕ is convex (resp. concave), the DM is said to be pessimistic (resp. optimistic). An inverse-S shaped transformation function is concave for small probabilities and convex for large probabilities. Wakker (2001) call such an inverse-S function as *cavex*. Wakker (2001) refer to such a function as *cavex*. Cavexity means that the function is initially concave and subsequently convex. Consequently, the DM exhibits optimism with respect to unlikely events and pessimism with respect to likely events, thereby placing relatively greater emphasis on extreme (best and worst) outcomes than on intermediate ones. The following graphs illustrate the three types of attitudes towards uncertainty.

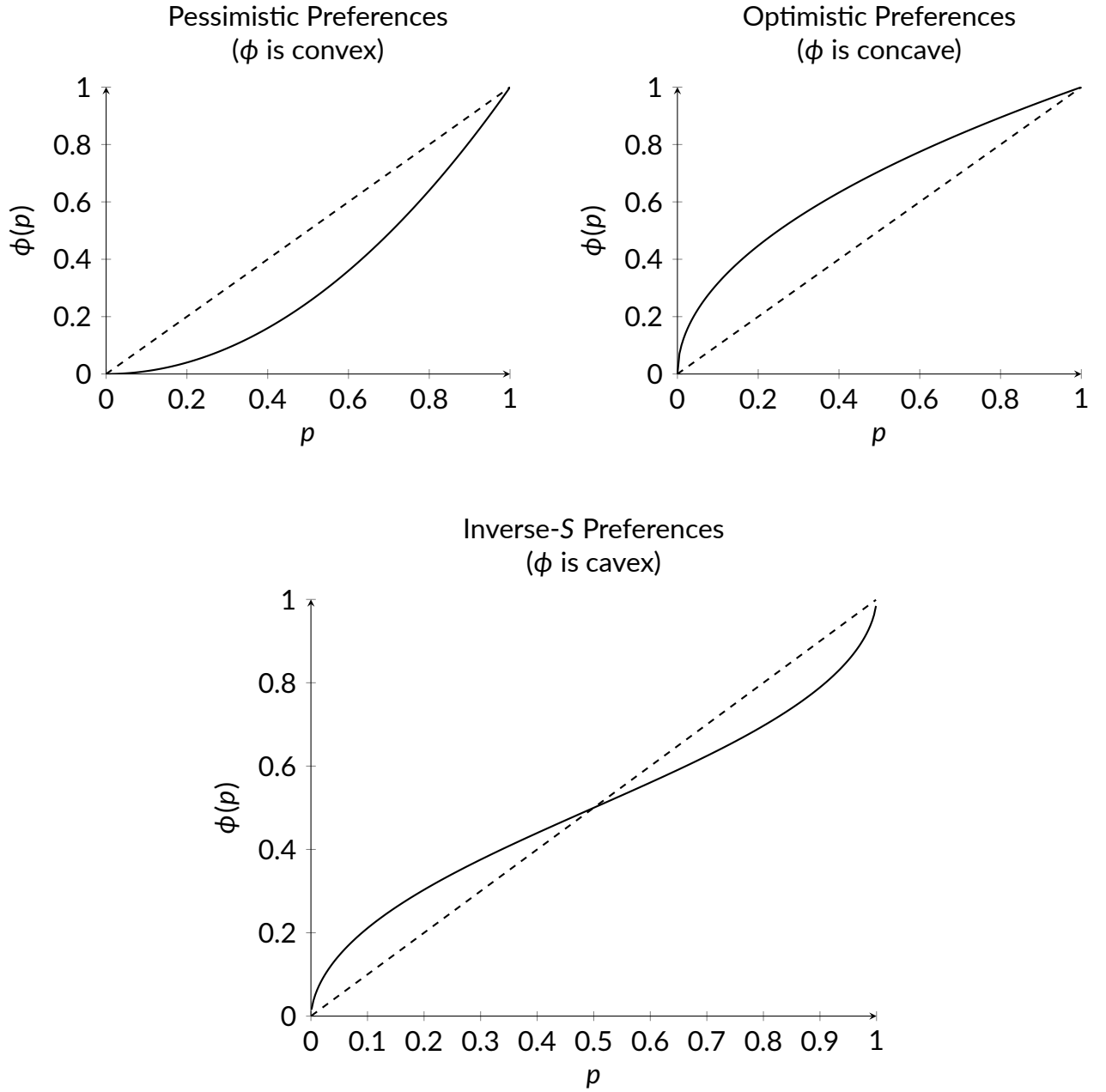


Figure 3: Pessimistic, Optimistic, and Inverse-S Attitudes towards Uncertainty.

7 Concluding Remarks

In this paper, we propose a simple updating rule for probability distortions and provide dynamic preferences foundations. The two key axioms are two-outcome dynamic consistency and conditional rank-dependent likelihood consistency. The approach taken in this paper suggests a new and natural procedure for updating capacities by the application of Bayes rule in situations where capacities can be expressed as probability distortions. The idea is as

follows. Under certain conditions, capacities can be expressed as transformations of additive probability measures, i.e. $\nu = \phi \circ p$. Two-outcome dynamic consistency ensures that the additive probability measure can be updated by using Bayes rule, i.e. $p_E(\cdot) = \frac{p(\cdot \cap E)}{p(E)}$ where $p(E) > 0$. By setting the updated capacity of each event $\nu_E(\cdot)$ equal to the transformation of the updated additive probability of the same event, $\phi_E(p_E(\cdot))$, we can obtain the corresponding capacity $\nu_E(\cdot) = \phi_E(p_E(\cdot))$. Conditional rank-dependent likelihood consistency ensures that the probability transformation function ϕ_E is unique, that is, $\phi_E = \phi$. This transformed function ϕ captures the DM's attitudes towards uncertainty: if the function ϕ is a convex (resp. concave), then this DM is pessimistic (resp. optimistic). In this sense, our simple updating rule for Choquet beliefs separates uncertainty attitudes and belief updating.

Appendix: Proofs

Proof of Proposition 1. Suppose $A \succcurlyeq_E^L B$. Thus,

$$\begin{aligned} A \succcurlyeq_E^L B &\iff \begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \notin A \end{bmatrix} \succcurlyeq_E \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \notin B \end{bmatrix} \\ &\iff \begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \in E \setminus A \\ y & \text{if } s \notin E \end{bmatrix} \succcurlyeq_E \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \in E \setminus B \\ y & \text{if } s \notin E \end{bmatrix}. \end{aligned}$$

By conditional preference, we know that

$$\begin{aligned} \begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \in E \setminus A \\ y & \text{if } s \notin E \end{bmatrix} &\sim_E \begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \in E \setminus A \\ g(s) & \text{if } s \notin E \end{bmatrix} \quad \text{and} \\ \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \in E \setminus B \\ y & \text{if } s \notin E \end{bmatrix} &\sim_E \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \in E \setminus B \\ g(s) & \text{if } s \notin E \end{bmatrix} \end{aligned}$$

By transitivity of $\succsim_E$, we have

$$A \succsim_E^L B \implies \begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \in E \setminus A \\ g(s) & \text{if } s \notin E \end{bmatrix} \succsim_E \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \in E \setminus B \\ g(s) & \text{if } s \notin E \end{bmatrix}.$$

Suppose now $\begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \in E \setminus A \\ g(s) & \text{if } s \notin E \end{bmatrix} \succsim_E \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \in E \setminus B \\ g(s) & \text{if } s \notin E \end{bmatrix}$. By conditional preference, we know that

$$\begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \in E \setminus A \\ g(s) & \text{if } s \notin E \end{bmatrix} \sim_E \begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \in E \setminus A \\ y & \text{if } s \notin E \end{bmatrix} = \begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \notin A \end{bmatrix} \quad \text{and}$$

$$\begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \in E \setminus B \\ g(s) & \text{if } s \notin E \end{bmatrix} \sim_E \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \in E \setminus B \\ y & \text{if } s \notin E \end{bmatrix} = \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \notin B \end{bmatrix}.$$

By transitivity of $\succsim_E$, we have

$$\begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \notin A \end{bmatrix} \succsim_E \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \notin B \end{bmatrix} \implies A \succsim_E^L B.$$

□

Proof of Proposition 2. Suppose $\begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \notin A \end{bmatrix} \succsim \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \notin B \end{bmatrix}$. By Axiom 3 (Two-

Outcome Dynamic Consistency),

$$\begin{aligned} & \begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \in E \setminus A \\ y & \text{if } s \in S \setminus E \end{bmatrix} \succcurlyeq \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \in E \setminus B \\ y & \text{if } s \in S \setminus E \end{bmatrix} \\ \xLeftrightarrow{\text{Axiom 3}} & \begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \in E \setminus A \\ y & \text{if } s \in S \setminus E \end{bmatrix} \succcurlyeq_E \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \in E \setminus B \\ y & \text{if } s \in S \setminus E \end{bmatrix}. \end{aligned}$$

By Axiom 2 (Conditional Preference),

$$\begin{aligned} & \begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \in E \setminus A \\ y & \text{if } s \in S \setminus E \end{bmatrix} \sim_E \begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \in E \setminus A \\ x & \text{if } s \in S \setminus E \end{bmatrix} = \begin{bmatrix} x & \text{if } s \in A \cup (S \setminus E) \\ y & \text{if } s \in E \setminus A \end{bmatrix}; \\ & \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \in E \setminus B \\ y & \text{if } s \in S \setminus E \end{bmatrix} \sim_E \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \in E \setminus B \\ x & \text{if } s \in S \setminus E \end{bmatrix} = \begin{bmatrix} x & \text{if } s \in B \cup (S \setminus E) \\ y & \text{if } s \in E \setminus B \end{bmatrix}. \end{aligned}$$

By transitivity of the conditional preference $\succcurlyeq_E$,

$$\begin{bmatrix} x & \text{if } s \in A \cup (S \setminus E) \\ y & \text{if } s \in E \setminus A \end{bmatrix} \succcurlyeq_E \begin{bmatrix} x & \text{if } s \in B \cup (S \setminus E) \\ y & \text{if } s \in E \setminus B \end{bmatrix}.$$

By Axiom 3 (Two-Outcome Dynamic Consistency),

$$\begin{aligned} & \begin{bmatrix} x & \text{if } s \in A \cup (S \setminus E) \\ y & \text{if } s \in E \setminus A \end{bmatrix} \succcurlyeq_E \begin{bmatrix} x & \text{if } s \in B \cup (S \setminus E) \\ y & \text{if } s \in E \setminus B \end{bmatrix} \\ \xLeftrightarrow{\text{Axiom 3}} & \begin{bmatrix} x & \text{if } s \in A \cup (S \setminus E) \\ y & \text{if } s \in E \setminus A \end{bmatrix} \succcurlyeq \begin{bmatrix} x & \text{if } s \in B \cup (S \setminus E) \\ y & \text{if } s \in E \setminus B \end{bmatrix}. \end{aligned}$$

Define $C := S \setminus E$. Thus, Axiom P2* (Two-Outcome Sure-Thing Principle) must hold:

$$\begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \notin A \end{bmatrix} \succcurlyeq \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \notin B \end{bmatrix} \iff \begin{bmatrix} x & \text{if } s \in A \cup C \\ y & \text{if } s \notin A \cup C \end{bmatrix} \succcurlyeq \begin{bmatrix} x & \text{if } s \in B \cup C \\ y & \text{if } s \notin B \cup C \end{bmatrix}.$$

□

Proof of Theorem 1

Proof of Sufficiency

Definition 4 (Cancellation). A subjective likelihood order $\succcurlyeq^L$ satisfies *cancellation* if for all $A, B, A \cup C, B \cup C \in \mathcal{E}$ and $A \cap C = B \cap C = \emptyset$,

$$A \succcurlyeq^L B \iff A \cup C \succcurlyeq^L B \cup C.$$

It states that offering the DM a consolation prize in the event that neither A nor B occurs, but C does, has no effect on her guess.

Lemma 1. *The subjective likelihood order $\succcurlyeq^L$ over $\mathcal{E}$ satisfies cancellation.*

Proof of Lemma 1. We want to show that, if C is disjoint from A and B , then $A \succcurlyeq^L B$ if and only if $A \cup C \succcurlyeq^L B \cup C$. We assume $x \succ y$ and suppose that:

$$\begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \notin A \end{bmatrix} \succcurlyeq \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \notin B \end{bmatrix},$$

which is equivalent to our definition $A \succcurlyeq^L B$. The next trick is to rewrite each gamble as follows:

$$\begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \in C \\ y & \text{if } s \notin A \cup C \end{bmatrix} \succcurlyeq \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \in C \\ y & \text{if } s \notin B \cup C \end{bmatrix}.$$

The two gambles above both have the same outcome y on the event C . So now, we can apply the two-outcome sure-thing principle which says that preferences over gambles are

unaffected by changing outcomes that occur in the same event. Now we can simply replace y with x on the event C , so the preference remains the same, and the following must be true:

$$\begin{bmatrix} x & \text{if } s \in A \\ x & \text{if } s \in C \\ y & \text{if } s \notin A \cup C \end{bmatrix} \succsim \begin{bmatrix} x & \text{if } s \in B \\ x & \text{if } s \in C \\ y & \text{if } s \notin B \cup C \end{bmatrix}.$$

The above can be rewritten as:

$$\begin{bmatrix} x & \text{if } s \in A \cup C \\ y & \text{if } s \notin A \cup C \end{bmatrix} \succsim \begin{bmatrix} x & \text{if } s \in B \cup C \\ y & \text{if } s \notin B \cup C \end{bmatrix},$$

which is equivalent to $A \cup C \succsim^L B \cup C$. □

Definition 5 (Ordinal Additivity). The capacity ν is *ordinal additive* if for all $A, B, A \cup C, B \cup C \in \mathcal{E}$ and $A \cap C = B \cap C = \emptyset$,

$$\nu(A) \geq \nu(B) \iff \nu(A \cup C) \geq \nu(B \cup C).$$

Lemma 2. *The capacity ν is ordinal additive.*

Proof of Lemma 2. By cancellation, we know that

$$A \succsim^L B \iff A \cup C \succsim^L B \cup C.$$

So, it is immediate that

$$\nu(A) \geq \nu(B) \iff \nu(A \cup C) \geq \nu(B \cup C).$$

□

Because $\succsim^L$ satisfies *cancellation*, it is a *qualitative probability* (Savage, 1954, p. 32). Now we turn to the characterisation of probabilistic sophistication for the context of CEU where we relate capacities to probabilities in the sense that capacities are transforms of probabilities. To proceed, two additional definitions are needed.

Definition 6 (Solvability). For all events $A \subset C$ and $v(A) \leq \beta \leq v(C)$, there exists an event B such that $A \subset B \subset C$ and $v(B) = \beta$.

This condition was introduced by Gilboa (1987) under the name convex-rangedness.

Definition 7 (Set-Continuity). If $A_n \uparrow A$ (i.e. $A_{n+1} \supset A_n$ and $\cup A_n = A$), then $\lim_{i \rightarrow \infty} v(A_i) = v(A)$;

Given a monotone increasing sequence of events $A_n \uparrow A$ (that is, a sequence $A_1 \subset A_2 \subset \dots$ converging to A), the limit of v of A_i as A_i approaches ∞ equals A .

The following result is provided by Theorem 8.1 of Sarin and Wakker (1998b), which we summarise in the following lemma.

Lemma 3. Let $\mathcal{E}$ be a σ -algebra and let CEU hold. There exists a countably additive atomless probability measure p and a strictly increasing continuous transformation ϕ such that $v = \phi \circ p$ if and only if the following conditions hold:

- (i) v is ordinal additive.
- (ii) v satisfies solvability;
- (iii) Set-continuity holds;

Proof of Lemma 3. First assume that the conditions (i), (ii) and (iii) hold. We prove the existence of p and ϕ as described in the theorem. We write $A \succsim^L B$ for events A, B whenever $v(A) \geq v(B)$. No confusion with the preference relation $\succsim$ on gambles will arise. First, we derive the properties of a qualitative probability ordering for $\succsim^L$.

Obviously, $\succsim^L$ is a weak order; the notation $\succ^L$ is as usual. We have $S \succsim^L A \succsim^L \emptyset$ for all events A and $S \succ \emptyset$. Finally, assume that event C is disjoint from events A and B . Then, the following four statements are equivalent:

- (1) $A \succsim^L B$;
- (2) $v(A) \geq v(B)$;
- (3) $v(A \cup C) \geq v(B \cup C)$;
- (4) $A \cup C \succsim^L B \cup C$.

The equivalence of (1) and (4) is the well-known additivity condition of qualitative probability theory. Thus, $\succsim$ is a qualitative probability ordering (Villegas, 1964).

Solvability of ν implies that no atoms exist and Condition (iii) implies monotone continuity of Villegas (1964). Therefore, by Villegas's Theorem 4.3, there exists a unique countably additive atomless probability measure p on the σ -algebra of events that represents the qualitative probability ordering $\succsim^L$. Hence, there exists a strictly increasing transformation ϕ such that $\nu = \phi \circ p$. By solvability of ν , the range of ϕ is $[0, 1]$, hence ϕ must be continuous.

For the reversed implication, ordinal additivity follows from additivity of p . Solvability of ν is implied because p is atomless and ϕ is continuous. Condition (iii) is implied by σ -additivity of p and continuity of ϕ . $\square$

The above proof is based on the qualitative probability result of Villegas (1964), which implies countably additivity measures. Finite additivity can be obtained by using different probability results, such as Savage (1954). Then, however, the fineness condition is more complicated (Gilboa, 1985).

So far, we have obtained that $\nu = \phi \circ p$. Next, we want to show that $\nu_E(A) = \phi_E \circ p_E(A)$, where $p_E(A) = \frac{p(A \cap E)}{p(E)}$. The approach taken in this paper suggests a simple procedure for updating capacities by the Bayes' rule. Under the conditions specified above, a capacity ν can be expressed as the transformation ϕ of an additive probability measure p . That is, $\nu = \phi \circ p$ with the transformation ϕ that is strictly increasing everywhere. First, we update the additive probability measure by using the Bayes' rule. Then, we update the corresponding capacity by setting the capacity of each event equal to the transformation of the updated additive probability of the same event. Formally, for each event $A \in \mathcal{E}$, the updated capacity ν_E is

$$\nu_E(A) = \phi_E(p_E(A)),$$

$$\text{where } p_E(A) = \frac{p(A \cap E)}{p(E)}.$$

Lemma 4. Assume Axioms 1–3, the unconditional likelihood order $\succsim^L$ over $\mathcal{E}$ represented by p , and the conditional likelihood order $\succsim_E^L$ represented by $\nu_E = \phi_E \circ p_E$. Thus, for each non-null event $E \in \mathcal{E}$ where $p(E) > 0$, there exists a strictly increasing, continuous transformation function ϕ_E

such that

$$v_E(\cdot) = \phi_E\left(\frac{p(\cdot \cap E)}{p(E)}\right).$$

Proof of Lemma 4. Suppose $A, B \subseteq E$. Then we have

$$\begin{aligned} A \succsim^L B &\iff v(A) \geq v(B) \iff \phi(p(A)) \geq \phi(p(B)) \iff p(A) \geq p(B) \\ &\iff \begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \notin A \end{bmatrix} \succsim \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \notin B \end{bmatrix} \\ &\xLeftrightarrow{\text{Axiom 3}} \begin{bmatrix} x & \text{if } s \in A \\ y & \text{if } s \notin A \end{bmatrix} \succsim_E \begin{bmatrix} x & \text{if } s \in B \\ y & \text{if } s \notin B \end{bmatrix} \\ &\iff A \succsim_E^L B \iff \phi_E(p_E(A)) \geq \phi_E(p_E(B)) \iff p_E(A) \geq p_E(B). \end{aligned}$$

Define $\tilde{p}(\cdot) := \frac{p(\cdot)}{p(E)}$ for all subsets of E . So, $\tilde{p}(E) = 1$, $\tilde{p}(\emptyset) = 0$ and $\tilde{p}(A) \geq \tilde{p}(B)$ if and only if $p_E(A) \geq p_E(B)$. So, $\tilde{p}(\cdot) = p_E(\cdot)$. Since $A \succsim^L B$ if and only if $p(A) \geq p(B)$, for each $\succsim_E$ we know $A \succsim_E^L B$ if and only if $p_E(A) \geq p_E(B)$, where $p_E(\cdot) = \frac{p(\cdot)}{p(E)}$.

Now consider the general case where $A, B \not\subseteq E$, and we want to show $p_E(A) = \frac{p(A \cap E)}{p(E)}$ for any event $A \in \mathcal{E}$. Define $f := x_{A \cap E}y$ and $g := x_A y$. Alternatively, f and g can be rewritten as

$$x_{A \cap E}y = \begin{bmatrix} x & \text{if } s \in A \cap E \\ y & \text{if } s \in A \setminus E \\ y & \text{if } s \in E \setminus A \\ y & \text{if } s \in S \setminus (A \cup E) \end{bmatrix} \quad \text{and} \quad x_A y = \begin{bmatrix} x & \text{if } s \in A \cap E \\ x & \text{if } s \in A \setminus E \\ y & \text{if } s \in E \setminus A \\ y & \text{if } s \in S \setminus (A \cup E) \end{bmatrix}.$$

By Axiom 2 (Conditional Preference), if $f = g$ on E then $f \sim_E g$. Notice that $x_{A \cap E}y = x_A y$ on E , so that $x_{A \cap E}y \sim_E x_A y$. By the definition of $\succsim_E^L$, $A \sim_E^L A \cap E$. Similarly, $B \sim_E^L B \cap E$. By transitivity of Axiom 1 (Weak Ordering), if $A \succsim_E^L B$ then $A \cap E \succsim_E^L B \cap E$. In other words, the conditional likelihood order $\succsim_E^L$ is defined only over the subset of events $\mathcal{E}_E := \{H \in \mathcal{E} : H \subseteq E\}$. If $A, B \in \mathcal{E}_E$, then $A \cap E = A$ and $B \cap E = B$. The unconditional likelihood order $\succsim^L$ over $\mathcal{E}$ is represented by the probability measure p defined over $\mathcal{E}$. Axiom 3 (Two-Outcome Dynamic Consistency) means that for all $A, B \in \mathcal{E}_E$ we have $A \succsim^L B$ if and only if $A \succsim_E^L B$, so that

$\succsim_E^I$ can be represented by p as well. However, p is not a well-defined probability measure on $\mathcal{E}_E$ because $p(E) < 1$, but we can define $\tilde{p}(\cdot) = \frac{p(\cdot)}{p(E)}$ over $\mathcal{E}_E$. We can define it for all events $A \in \mathcal{E}$ by letting $p_E(A) = \tilde{p}(A \cap E)$. p_E is thus a probability measure on $\mathcal{E}$, e.g. $p_E(S) = \tilde{p}(S \cap E) = \tilde{p}(E) = 1$. So,

$$\begin{aligned} A \cap E \succsim_E^I B \cap E &\iff \tilde{p}(A \cap E) \geq \tilde{p}(B \cap E) \\ &\iff \frac{p(A \cap E)}{p(E)} \geq \frac{p(B \cap E)}{p(E)} \\ &\iff \phi_E\left(\frac{p(A \cap E)}{p(E)}\right) \geq \phi_E\left(\frac{p(B \cap E)}{p(E)}\right). \end{aligned}$$

□

So far, we have obtained that $v_E = \phi_E \circ p_E$. Next, we want to show that $v_E(\cdot) = \phi \circ p_E(\cdot)$, where $p_E(\cdot) = \frac{p(\cdot \cap E)}{p(E)}$. Recall that we have the capacity conditional on each non-null event $E \in \mathcal{E}$ with $v_E(A) = \phi_E(p_E(A))$. Now suppose that we can match subjective acts with lotteries. Let $u(\bar{x}) = 1$ for the best outcome and $u(\underline{x}) = 0$ for the worst outcome. To capture capacities, we match them with probabilities. Suppose that we can find a probability $\alpha \in [0, 1]$ such that

$$\begin{aligned} (A, \bar{x}; \text{not } A, \underline{x}) &\sim (\alpha, \bar{x}; 1 - \alpha, \underline{x}); \\ (\tilde{A}, \bar{x}; \text{not } \tilde{A}, \underline{x}) &\sim_E (\alpha, \bar{x}; 1 - \alpha, \underline{x}). \end{aligned} \tag{8}$$

This lottery matches the subjective acts in both unconditional and conditional scenarios.

If the DM is a CEU maximiser and capacities are transforms of probabilities, then the two indifferences in (8) are equivalent to:

$$\begin{aligned} \phi(p(A))u(\bar{x}) + (1 - \phi(p(A)))u(\underline{x}) &= \phi(\alpha)u(\bar{x}) + (1 - \phi(\alpha))u(\underline{x}); \\ \phi_E(p_E(\tilde{A}))u(\bar{x}) + (1 - \phi_E(p_E(\tilde{A})))u(\underline{x}) &= \phi_E(\alpha)u(\bar{x}) + (1 - \phi_E(\alpha))u(\underline{x}). \end{aligned}$$

Since $u(\bar{x}) = 1$ and $u(\underline{x}) = 0$, it must be

$$\phi(p(A)) = \phi(\alpha) \quad \text{and} \quad \phi_E(p_E(\tilde{A})) = \phi_E(\alpha).$$

Hence, the DM reveals that $p(A) = \alpha$ and $p_E(\tilde{A}) = \alpha$, so that $p(A) = p_E(\tilde{A})$. Similarly, we can obtain $p(A') = p_E(\tilde{A}')$, $p(B) = p_E(\tilde{B})$ and $p(B') = p_E(\tilde{B}')$.

Now we could find $A, B, A', B', \tilde{A}, \tilde{B}, \tilde{A}', \tilde{B}' \in \mathcal{E}$ such that

1. $p(A) = p_E(\tilde{A}) =: \alpha$;
2. $p(A') = p_E(\tilde{A}') =: \alpha'$;
3. $p(B) = p_E(\tilde{B}) =: \beta$;
4. $p(B') = p_E(\tilde{B}') =: \beta'$.

Lemma 5. Assume Axiom 4. For each conditional preference relation $\succsim_E$ and any non-null event $E \in \mathcal{E}$, we have a uniform probability transformation function, $\phi_E = \phi$.

Proof of Lemma 5. In Axiom 4, the first pair of the two unconditional indifferences (i.e. $A_i F \sim B_i G$ and $A'_i F \sim B'_i G$) imply $\phi(\alpha) - \phi(\alpha') = \phi(\beta) - \phi(\beta')$, and the second pair of the conditional indifferences (i.e. $\tilde{A}_j Y \sim_E \tilde{B}_j Z$ and $\tilde{A}'_j Y \sim_E \tilde{B}'_j Z$) imply $\phi_E(\alpha) - \phi_E(\alpha') = \phi_E(\beta) - \phi_E(\beta')$. So, this type of likelihood consistency condition holds if ϕ and ϕ_E order differences in the same way. That is,

$$\phi(\alpha) - \phi(\alpha') = \phi(\beta) - \phi(\beta') \iff \phi_E(\alpha) - \phi_E(\alpha') = \phi_E(\beta) - \phi_E(\beta').$$

If so, then these functions must be positive affine transformations: $\phi = \lambda\phi_E + \gamma$, where $\lambda > 0$ and $\gamma \in \mathbb{R}$. Note that $\lambda = 1$ follows from $\phi(1) = \phi_E(1) = 1$ and $\gamma = 0$ follows from $\phi(0) = \phi_E(0) = 0$, so that we obtain $\phi = \phi_E$. $\square$

Proof of Necessity

Suppose that our proposed simple updating model for Choquet beliefs holds. That is,

$$f \succsim_E g \iff \int_S u(f(s)) d\phi(p_E(s)) \geq \int_S u(g(s)) d\phi(p_E(s)). \quad (9)$$

It is immediate that completeness and transitivity hold so that Axiom 1 (Weak Ordering) holds.

Now, we will show that our simple representation result (9) implies Axiom 2 (Conditional Preference). If $f(s) = g(s)$ for each state s in any non-null event E , then we have $\int_S u(f(s)) d\phi(p_E(s)) = \int_S u(g(s)) d\phi(p_E(s))$, which implies $f \sim_E g$. So, Axiom 2 (Conditional Preference) holds.

Now, we will show that our simple representation result (9) implies Axiom 3 (Two-Outcome Dynamic Consistency). If $A, B \subseteq E$ and $x \succ y$, we can represent two-outcome gambles as follows:

$$\begin{aligned} x_A y \succcurlyeq x_B y &\iff \phi(p(A))u(x) + [1 - \phi(p(A))]u(y) \geq \phi(p(B))u(x) + [1 - \phi(p(B))]u(y) \\ &\iff [\phi(p(A)) - \phi(p(B))]u(x) \geq [\phi(p(A)) - \phi(p(B))]u(y). \end{aligned}$$

Since $x \succ y$ implies $u(x) > u(y)$, it must be either $\phi(p(A)) > \phi(p(B))$ or $\phi(p(A)) = \phi(p(B))$. Because $\phi(\cdot)$ is a strictly increasing function, it follows that either $p(A) > p(B)$ or $p(A) = p(B)$. Since the event $E \in \mathcal{E}$ is non-null, it must be $p(E) > 0$. Thus, dividing both sides by $p(E)$ yields $\frac{p(A)}{p(E)} > \frac{p(B)}{p(E)}$ or $\frac{p(A)}{p(E)} = \frac{p(B)}{p(E)}$, which follows that either $\phi\left(\frac{p(A)}{p(E)}\right) > \phi\left(\frac{p(B)}{p(E)}\right)$ or $\phi\left(\frac{p(A)}{p(E)}\right) = \phi\left(\frac{p(B)}{p(E)}\right)$. In other words, $\phi\left(\frac{p(A)}{p(E)}\right) - \phi\left(\frac{p(B)}{p(E)}\right) \geq 0$. Since $u(x) > u(y)$, we have

$$\begin{aligned} &\left[\phi\left(\frac{p(A)}{p(E)}\right) - \phi\left(\frac{p(B)}{p(E)}\right)\right]u(x) \geq \left[\phi\left(\frac{p(A)}{p(E)}\right) - \phi\left(\frac{p(B)}{p(E)}\right)\right]u(y) \\ &\iff [\phi(p_E(A)) - \phi(p_E(B))]u(x) \geq [\phi(p_E(A)) - \phi(p_E(B))]u(y) \\ &\iff \phi(p_E(A))u(x) + [1 - \phi(p_E(A))]u(y) \geq \phi(p_E(B))u(x) + [1 - \phi(p_E(B))]u(y) \\ &\implies x_A y \succcurlyeq_E x_B y. \end{aligned}$$

Therefore, Axiom 3 (Two-Outcome Dynamic Consistency) holds.

Finally, we will show that our simple representation result (9) implies Axiom 4 (Conditional Rank-Dependent Likelihood Consistency). Recall the definition of the decumulative gamble, e.g. F . The indifference of decumulative gambles $A_i F \sim B_i G$ can be expressed as the following:

$$(F_1, \dots, F_{i-1}, A, F_{i+1}, \dots, F_n) \sim (G_1, \dots, G_{i-1}, B, G_{i+1}, \dots, F_n). \quad (10)$$

Using the CEU evaluation function, the indifference in (10) yields:

$$\begin{aligned}
& u(x_0) + v(A) \left(u(x_i) - u(x_{i-1}) \right) + \sum_{\substack{j=1 \\ j \neq i}}^n v(F_j) \left(u(x_j) - u(x_{j-1}) \right) \\
& = u(x_0) + v(B) \left(u(x_i) - u(x_{i-1}) \right) + \sum_{\substack{j=1 \\ j \neq i}}^n v(G_j) \left(u(x_j) - u(x_{j-1}) \right).
\end{aligned} \tag{11}$$

Similarly, the indifference of decumulative gambles $A'_i F \sim B'_i G$ can be expressed as the following:

$$(F_1, \dots, F_{i-1}, A', F_{i+1}, \dots, F_n) \sim (G_1, \dots, G_{i-1}, B', G_{i+1}, \dots, G_n). \tag{12}$$

Using the CEU evaluation function, the indifference in (12) yields:

$$\begin{aligned}
& u(x_0) + v(A') \left(u(x_i) - u(x_{i-1}) \right) + \sum_{\substack{j=1 \\ j \neq i}}^n v(F_j) \left(u(x_j) - u(x_{j-1}) \right) \\
& = u(x_0) + v(B') \left(u(x_i) - u(x_{i-1}) \right) + \sum_{\substack{j=1 \\ j \neq i}}^n v(G_j) \left(u(x_j) - u(x_{j-1}) \right).
\end{aligned} \tag{13}$$

The indifference of decumulative gambles $\tilde{A}_i Y \sim_E \tilde{B}_i Z$ can be expressed as the following:

$$(Y_1, \dots, Y_{i-1}, \tilde{A}, Y_{i+1}, \dots, Y_n) \sim_E (Z_1, \dots, Z_{i-1}, \tilde{B}, Z_{i+1}, \dots, Z_n). \tag{14}$$

Using the CEU evaluation function, the indifference in (14) yields:

$$\begin{aligned}
& u(x_0) + v_E(\tilde{A}) \left(u(x_i) - u(x_{i-1}) \right) + \sum_{\substack{j=1 \\ j \neq i}}^n v_E(Y_j) \left(u(x_j) - u(x_{j-1}) \right) \\
& = u(x_0) + v_E(\tilde{B}) \left(u(x_i) - u(x_{i-1}) \right) + \sum_{\substack{j=1 \\ j \neq i}}^n v_E(Z_j) \left(u(x_j) - u(x_{j-1}) \right).
\end{aligned} \tag{15}$$

Since $v_E(\cdot) = \phi(p_E(\cdot))$, where $\phi(\cdot)$ is a strictly increasing function, we have the following:

$$\begin{aligned} & u(x_0) + \phi(p_E(\tilde{A})) \left(u(x_i) - u(x_{i-1}) \right) + \sum_{\substack{j=1 \\ j \neq i}}^n \phi(p_E(Y_j)) \left(u(x_j) - u(x_{j-1}) \right) \\ &= u(x_0) + \phi(p_E(\tilde{B})) \left(u(x_i) - u(x_{i-1}) \right) + \sum_{\substack{j=1 \\ j \neq i}}^n \phi(p_E(Z_j)) \left(u(x_j) - u(x_{j-1}) \right). \end{aligned}$$

We also assume $A \sim^* \tilde{A}|E$, $A' \sim^* \tilde{A}'|E$, $B \sim^* \tilde{B}|E$ and $B' \sim^* \tilde{B}'|E$, so that we have the following indifference:

$$\begin{aligned} A \sim^* \tilde{A}|E &\iff \bar{x}_{A\underline{x}} \sim \bar{x}_{\alpha\underline{x}} \\ &\quad \bar{x}_{\tilde{A}\underline{x}} \sim_E \bar{x}_{\alpha\underline{x}} \\ &\implies \phi(p(A))u(\bar{x}) + \left(1 - \phi(p(A))\right)u(\underline{x}) = \phi(\alpha)u(\bar{x}) + (1 - \phi(\alpha))u(\underline{x}) \\ &\implies \phi(p_E(\tilde{A}))u(\bar{x}) + \left(1 - \phi(p_E(\tilde{A}))\right)u(\underline{x}) = \phi(\alpha)u(\bar{x}) + (1 - \phi(\alpha))u(\underline{x}) \\ &\implies \phi(p(A))u(\bar{x}) = \phi(\alpha)u(\bar{x}) \\ &\implies \phi(p_E(\tilde{A}))u(\bar{x}) = \phi(\alpha)u(\bar{x}) \\ &\implies \phi(p(A)) = \phi(\alpha) \implies p(A) = p_E(\tilde{A}) = \alpha; \\ &\quad \phi(p_E(\tilde{A})) = \phi(\alpha) \end{aligned}$$

Similarly, we can obtain:

$$\begin{aligned} A' \sim^* \tilde{A}'|E &\iff \bar{x}_{A'\underline{x}} \sim \bar{x}_{\alpha'\underline{x}} \implies p(A') = p_E(\tilde{A}') = \alpha'; \\ &\quad \bar{x}_{\tilde{A}'\underline{x}} \sim_E \bar{x}_{\alpha'\underline{x}} \\ B \sim^* \tilde{B}|E &\iff \bar{x}_{B\underline{x}} \sim \bar{x}_{\beta\underline{x}} \implies p(B) = p_E(\tilde{B}) = \beta; \\ &\quad \bar{x}_{\tilde{B}\underline{x}} \sim_E \bar{x}_{\beta\underline{x}} \\ B' \sim^* \tilde{B}'|E &\iff \bar{x}_{B'\underline{x}} \sim \bar{x}_{\beta'\underline{x}} \implies p(B') = p_E(\tilde{B}') = \beta'. \\ &\quad \bar{x}_{\tilde{B}'\underline{x}} \sim_E \bar{x}_{\beta'\underline{x}} \end{aligned}$$

Deducting Equation (13) from Equation (11) and cancelling common terms gives:

$$v(A) - v(A') = v(B) - v(B') \iff \phi(p(A)) - \phi(p(A')) = \phi(p(B)) - \phi(p(B')). \quad (16)$$

Since $p(A) = p_E(\tilde{A}) = \alpha$, $p(A') = p_E(\tilde{A}') = \alpha'$, $p(B) = p_E(\tilde{B}) = \beta$ and $p(B') = p_E(\tilde{B}') = \beta'$, the revealed difference/tradeoff (16) is equivalent to

$$\phi(p_E(\tilde{A})) - \phi(p_E(\tilde{A}')) = \phi(p_E(\tilde{B})) - \phi(p_E(\tilde{B}')) \iff v_E(\tilde{A}) - v_E(\tilde{A}') = v_E(\tilde{B}) - v_E(\tilde{B}'). \quad (17)$$

Equation (17) holds if Equation (15) and the following holds:

$$\begin{aligned} & u(x_0) + v_E(\tilde{A}') \left(u(x_i) - u(x_{i-1}) \right) + \sum_{\substack{j=1 \\ j \neq i}}^n v_E(Y_j) \left(u(x_j) - u(x_{j-1}) \right) \\ & = u(x_0) + v_E(\tilde{B}') \left(u(x_i) - u(x_{i-1}) \right) + \sum_{\substack{j=1 \\ j \neq i}}^n v_E(Z_j) \left(u(x_j) - u(x_{j-1}) \right). \end{aligned} \quad (18)$$

Equation (18) holds if and only if $(Y_1, \dots, Y_{i-1}, \tilde{A}', Y_{i+1}, \dots, Y_n) \sim_E (Z_1, \dots, Z_{i-1}, \tilde{B}', Z_{i+1}, \dots, Z_n)$; that is, $\tilde{A}'Y \sim_E \tilde{B}'Z$. Hence, Axiom 4 (Conditional Rank-Dependent Likelihood Consistency) holds. The proof of necessity is now complete.

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